

UGTS-KC 4.2

GENERAL OPERATOR AND ORDER ADDENDUM

Typed mathematical operator families, precedence, dependency, composition, reduction and event order

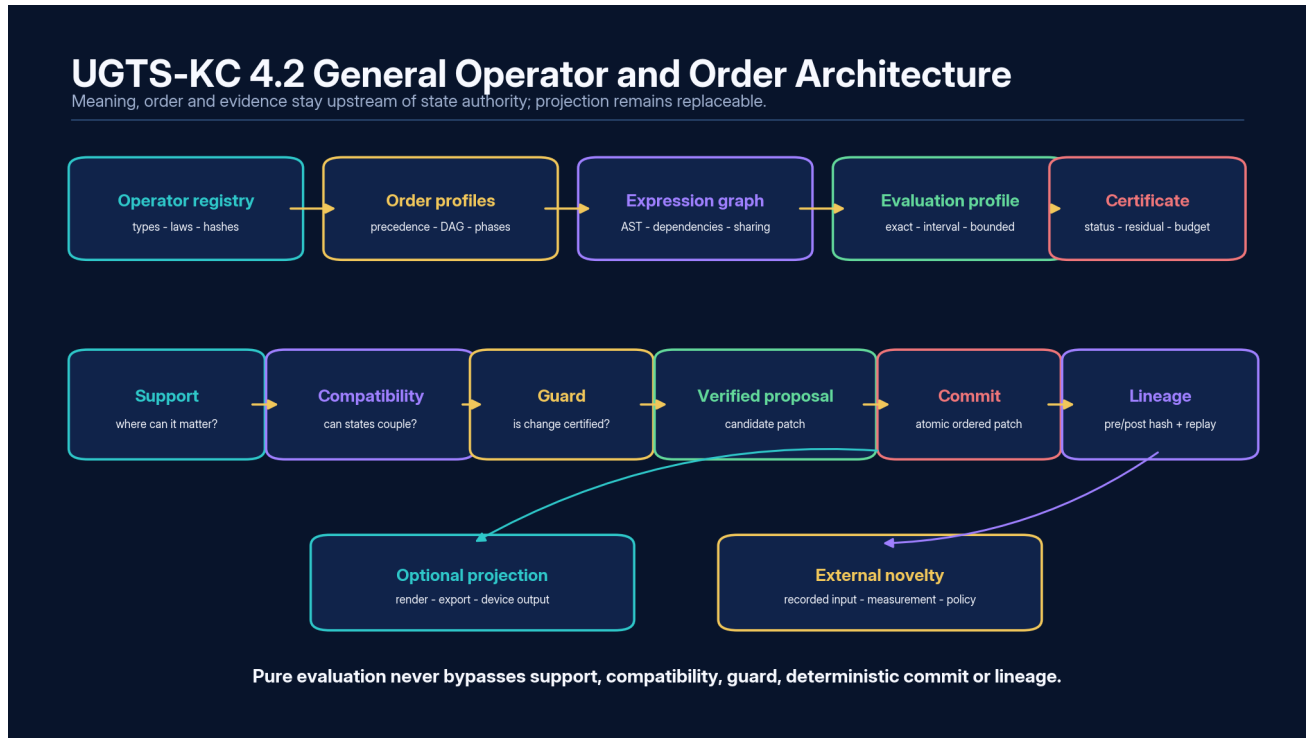


Figure 1. General operator meaning and evaluation remain upstream of the canonical UGTS state-authority chain.

Release field	Value
Prepared for	Tom Klootwijk
Version / date	4.2.0 / 28 August 2026
Codename	GOOA - General Operator and Order Addendum
New mechanism delta	M630-M885 (256 additions)
Catalog scope	16 domains x 16 operator/order mechanisms
Reference validation	441 tests PASS; wheel/sdist and clean install
Schema	ugts-kc-general-operator-order 4.2

ATTRIBUTION AND EVIDENCE BOUNDARY

Prepared for Tom Klootwijk. Requester-supplied identity and attribution data are not independently verified. This is a technical design and executable reference artifact, not legal proof of identity, ownership, priority or patentability. A finite catalog does not literally enumerate all future mathematics; it provides general typed operator-family and extension contracts.

Contents and Executive Summary

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Executive summary

UGTS-KC 4.2 turns mathematical operator meaning and every order that can affect a result into literal, versioned substrate data. It adds a general operator record, three precedence profiles, a canonical fifteen-phase schedule, exactness/effect/capability metadata, deterministic reduction and event-order rules, and an open extension namespace. The retained authority chain is unchanged: pure evaluation may support a state-changing proposal, but only support, compatibility, guard, verification, deterministic commit and lineage can alter authoritative state.

RELEASE DECISION

GO for a general addendum with a finite 256-entry catalog plus open operator-family generators. Reject the literal claim that a finite document can enumerate every mathematical operator. The released completeness claim is structural: every shipped operator has a typed record and every result-affecting order has an explicit profile.

Measured item	4.2 result	Meaning
Catalog	256 records, M630-M885	Sixteen major domains; sixteen entries per domain.
Implementation	194 executable / 62 specified	Advanced operators remain typed contracts rather than false implementation claims.
Orders	3 precedence profiles; 15 phases	Parse, dependency, composition, reduction, parallel, event and replay order are distinct.

Measured item	4.2 result	Meaning
Tests	441/441 PASS	Catalog rows, hashes, schema, expressions, intervals, numerics, events, CLI and builds.
Runtime	1626 Python source lines	Dependency-free standard-library reference subset.
Distributions	wheel + source archive	Clean-install CLI validation completed.

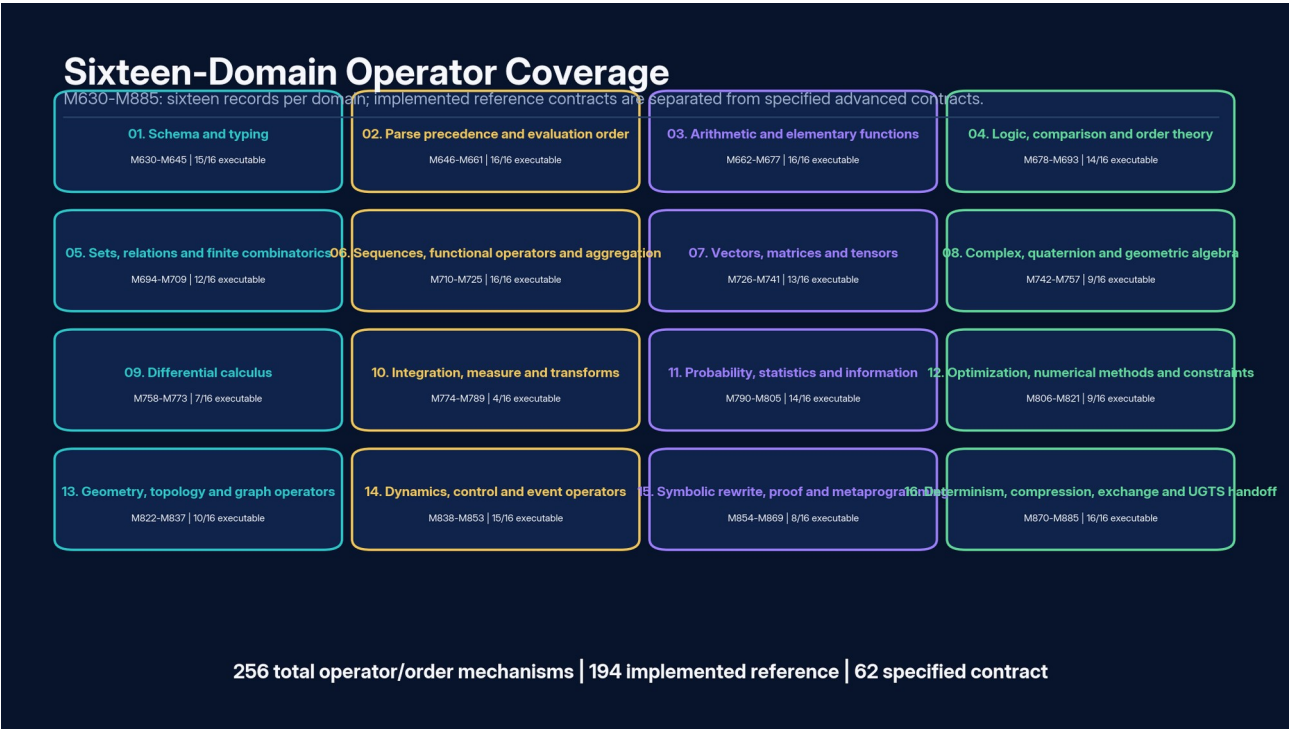


Figure 2. The 256-entry delta covers sixteen domains while visibly separating executable reference mechanisms from specified contracts.

1. Release Definition and Decision

1.1 What this version adds

Earlier UGTS reports already provide broad geometry, fields, topology, kinematics, dynamics, scene identity, deterministic proposal commit, replay, content-addressed definitions, interval guards, finite keys and self-contained runtime delivery. The missing general layer was a single record that can state what an operator means and a single order calculus that can state exactly how operators are grouped and executed across those domains.

- A content-addressed general OperatorSpec with syntax, arity, domain, codomain, units, shape, exactness, algebraic laws, effects, capabilities, dependencies, invariants, provenance and version.
- Explicit parser precedence, associativity and fixity profiles, including the unary-sign/power boundary and comparison-chain semantics.
- Distinct dependency, phase, composition, reduction, parallel, event and replay orders rather than one overloaded word order.
- A 256-entry engineering catalog covering arithmetic through symbolic proof and UGTS handoff.
- A bounded reference evaluator and certificate path that refuses arbitrary code and unsupported mathematical claims.

1.2 Formal release decision

DECISION - GO

Admit the general operator and order layer as UGTS-KC 4.2. Keep all mathematical results typed and evidence-bounded; keep state mutation proposal-based and event-authoritative; keep projection downstream. Treat “all possible operators” as an extensible schema objective, not a false finite-enumeration claim.

1.3 Catalog continuity

The project merged catalog spine reaches M629 in the retained 3.9.4 Bayer Direct continuity record. Version 4.2 appends M630-M885 without renumbering prior entries. The separate 4.0 and 4.1 spatial/native branches retain their own release namespaces and are referenced for observation, uncertainty, seed storage and promotion-gate discipline; this general addendum does not silently replace them.

2. Source Continuity and Evidence Discipline



Figure 3. The new layer generalizes retained definitions and order rules while preserving the canonical event sequence.

2.1 Immediate source basis

ID	Source	4.2 use	SHA-256 prefix
S1	UGTS-KC 3.9 KC Elizabeth Vector Game Runtime	deterministic game runtime, ordered phases, projects and validation	d7ee3c91abf60e35...
S2	UGTS-KC 3.6.2 SCLP	typed symbol separation, interval guards, finite indexing and evidence boundaries	6987490a0a7b7ed7...
S3	UGTS-KC 2.0 Expanded Substrate	patterns, fields, topology, kinematics, dynamics and event degeneracy	a29dd9959ebd300c...
S4	UGTS-KC Two Hands 3.0	stable scene identity, proposal verification, deterministic commit and replay	183ffa4e2bf29cfa...
S5	UGTS-GN 1.1 GPU-native Addendum	canonical authority chain, packed precision, compression and hardware evidence rules	3a6b6c57118ca4e2...
S6	UGTS-KC 3.6 Literal Referential Substrate	content-addressed definitions, dependency DAG and normative operation order	e21433cd8378368d...
S7	UGTS-KC 3.6.3 SARA	typed security/public boundaries and authorization guards	c5a8256e345047d5...

2.2 Preserved architectural commitments

Source commitment	General 4.2 treatment
Query-first authority	Operator evaluation never bypasses support, compatibility, guard, commit or lineage.
Definitions live inside substrate	Operator, profile and schedule records use stable IDs, dependencies and content hashes.
Finite typing and evidence boundary	Exact, bounded, approximate, stochastic, presentation-only and specified statuses remain distinct.
One bit has a narrow role	Flags, routes and masks never replace continuous values, uncertainty or history.
Compression requires equivalence	Packing, seed reconstruction and codebooks require declared decode/error semantics.
Projection is downstream	Render, export, storage, network and device adapters do not redefine mathematics or event state.

2.3 Evidence classes

Each M630-M885 row records a provenance class and validation state. “Implemented reference” means the package contains an executable bounded representative or directly checked record contract. “Specified contract” means the operator family is formally typed and integrated, but the dependency-free runtime does not claim a full production implementation. Neither status is independent scientific validation.

3. Formal UGTS-KC 4.2 Architecture

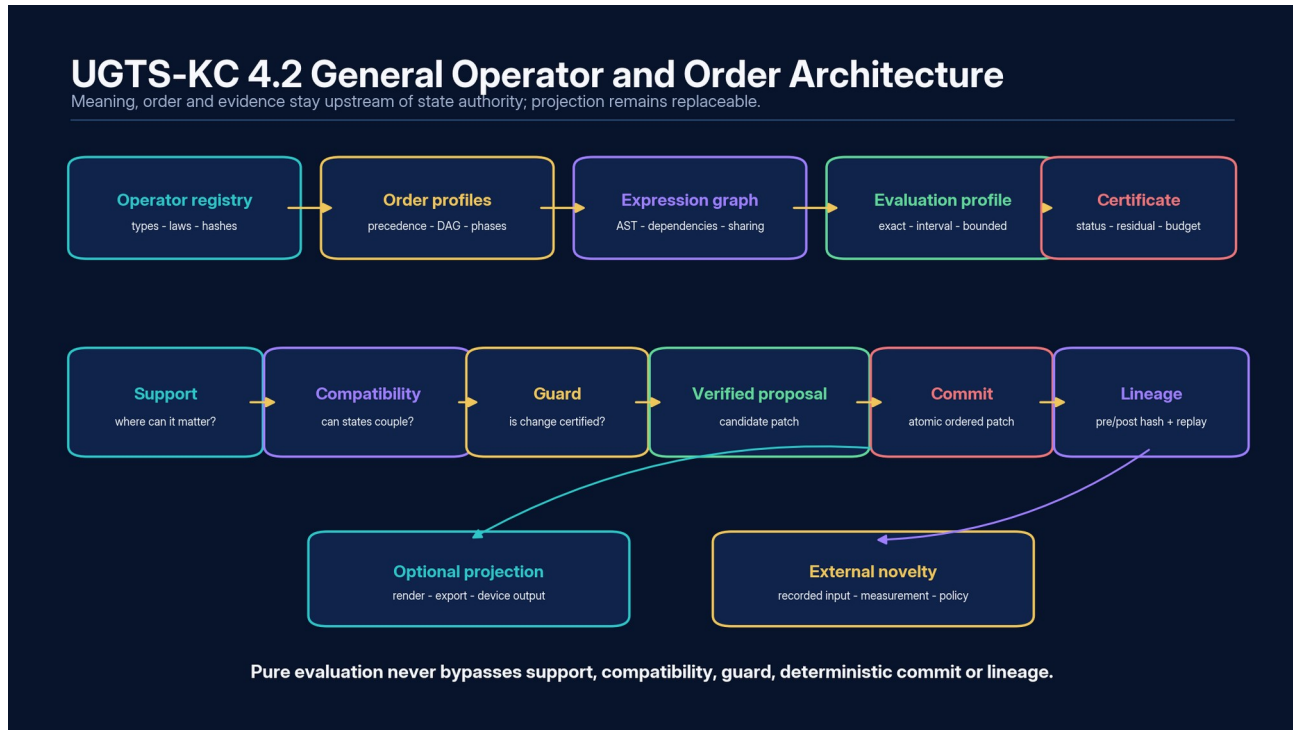


Figure 4. Operator registry, order profile, expression graph, evaluation profile and certificate feed the retained state-authority path.

3.1 Upgraded object

```

W_4.2 = W_retained + (0, Y, Q, A, B, Cert)

0    content-addressed operator registry
Y    type / sort / unit / shape universe
Q    precedence, dependency, phase, composition, reduction, event and replay orders
A    canonical expression and definition graph
B    finite work, precision, recursion and storage budgets
Cert exactness, residual, interval, status and provenance certificate
  
```

3.2 Canonical processing path

```

literal/operator definition
-> parse/group -> normalize -> resolve/hash -> type/shape/unit
-> scoped rewrite -> dependency/composition/reduction plan
-> exact or bounded evaluation -> certificate
-> support -> compatibility -> guard -> verified proposal
-> deterministic commit -> transition -> lineage/replay
-> optional projection/export/device adapter
  
```

3.3 Single source of operator truth

The same operator and order records are used for catalog validation, safe expression traces, example-project validation, deterministic hashes and the report appendices. No separate prose-only meaning is required to decide arity, grouping, exactness or authority status.

4. General Operator Record and Type Universe

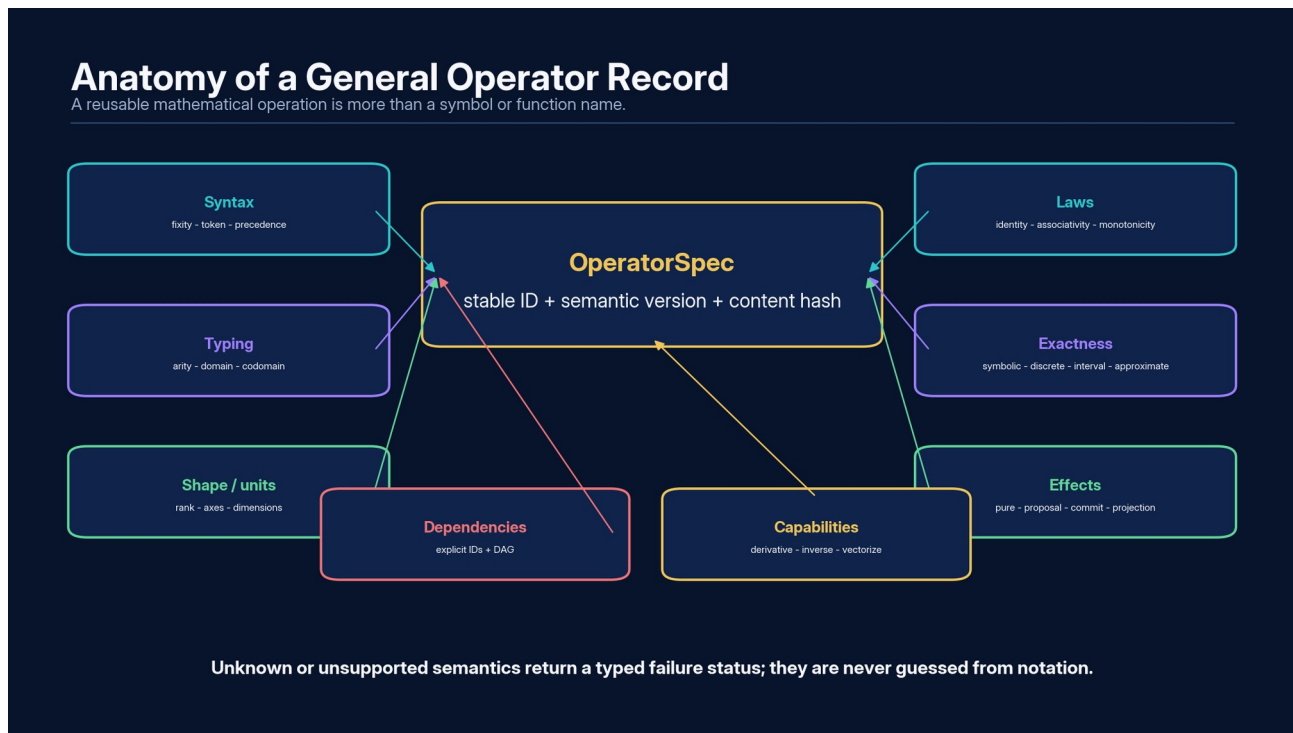


Figure 5. A general operator record separates syntax, typing, laws, exactness, effects, dependencies and capabilities.

4.1 Formal record

```

O_i = (id, version, syntax, arity, A_1..A_n, B, parameters,
      laws, exactness, effects, capabilities, dependencies,
      order_contract, invariants, provenance, content_hash)
  
```

Semantic arity is independent from notation. A binary operation may be written infix, functional or as an explicit AST node. A higher-order operator may consume or produce functions. A relation-valued operator may return a set or certificate rather than a single scalar. Partial and multivalued status is explicit.

4.2 Type families

Family	Representative sorts	Required contract
Scalars and numeric towers	integer, rational, real, complex, interval, modular, bit-vector	range, exactness, overflow and coercion
Collections	sequence, tuple, set, multiset, map, relation, stream	element type, cardinality/budget and order
Linear objects	vector, matrix, tensor, Clifford element	shape, rank, axes, field and contraction
Geometric/topological	point, frame, manifold chart, graph, cell complex	coordinate/gluing maps, orientation and invariants
Probability/information	event, distribution, random variable, estimator	normalization, seed, measure and uncertainty
State/event	state, proposal, transition, checkpoint, lineage	effects, authority and replay order

4.3 Laws are scoped data

Associativity, commutativity, identity, distributivity, idempotence, monotonicity, convexity and homomorphism properties are never inferred from a symbol alone. Matrix multiplication is associative but not commutative; floating-point addition is not treated as perfectly associative for deterministic reduction; interval and approximate equality are distinct from exact equality.

5. Order Calculus: Precedence, Dependencies and Phases

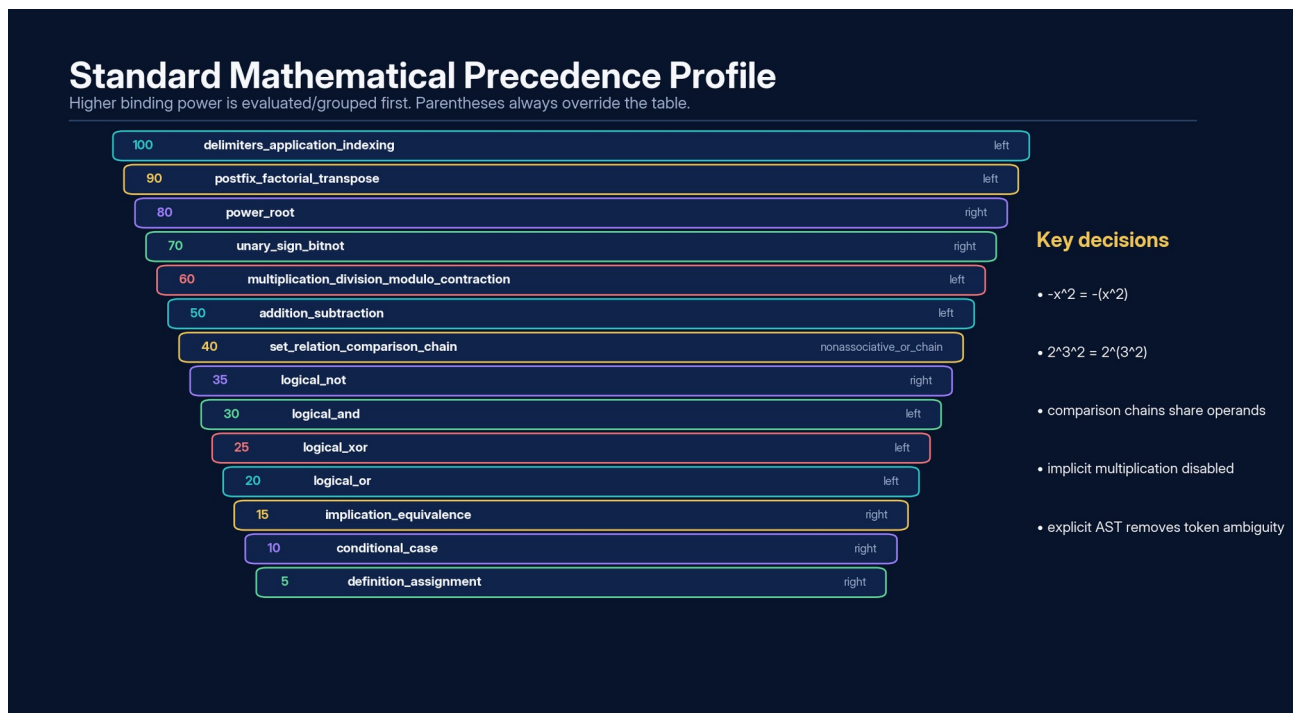


Figure 6. Standard mathematical precedence, with explicit decisions for unary signs, powers and implicit multiplication.

5.1 Parse order

The standard mathematical profile assigns binding powers to function application, postfix operations, power/root, unary operations, products, sums, comparisons, Boolean operators, conditionals and definitions. Parentheses override the profile. The executable Python-safe subset uses a restricted Python expression AST. The explicit-AST profile removes token precedence entirely for interchange.

5.2 Orders that must not be conflated

Order	Question answered	Failure if hidden
Precedence/associativity	How do tokens group?	Different AST and value.
Dependency	Which definitions must exist first?	Unknown reference or cycle.
Phase	When do typing, evaluation, guard and commit occur?	Pure work can mutate state or use invalid inputs.
Composition	Which map acts first?	Wrong frame, transform or function result.
Reduction	What tree/fold combines values?	Floating and non-associative drift.
Parallel	Which nodes may run concurrently and how do they rejoin?	Race-dependent output.
Event	Which simultaneous verified proposal wins or merges?	Silent overwrite.
Replay	In what sequence are patches checked?	State-hash divergence.

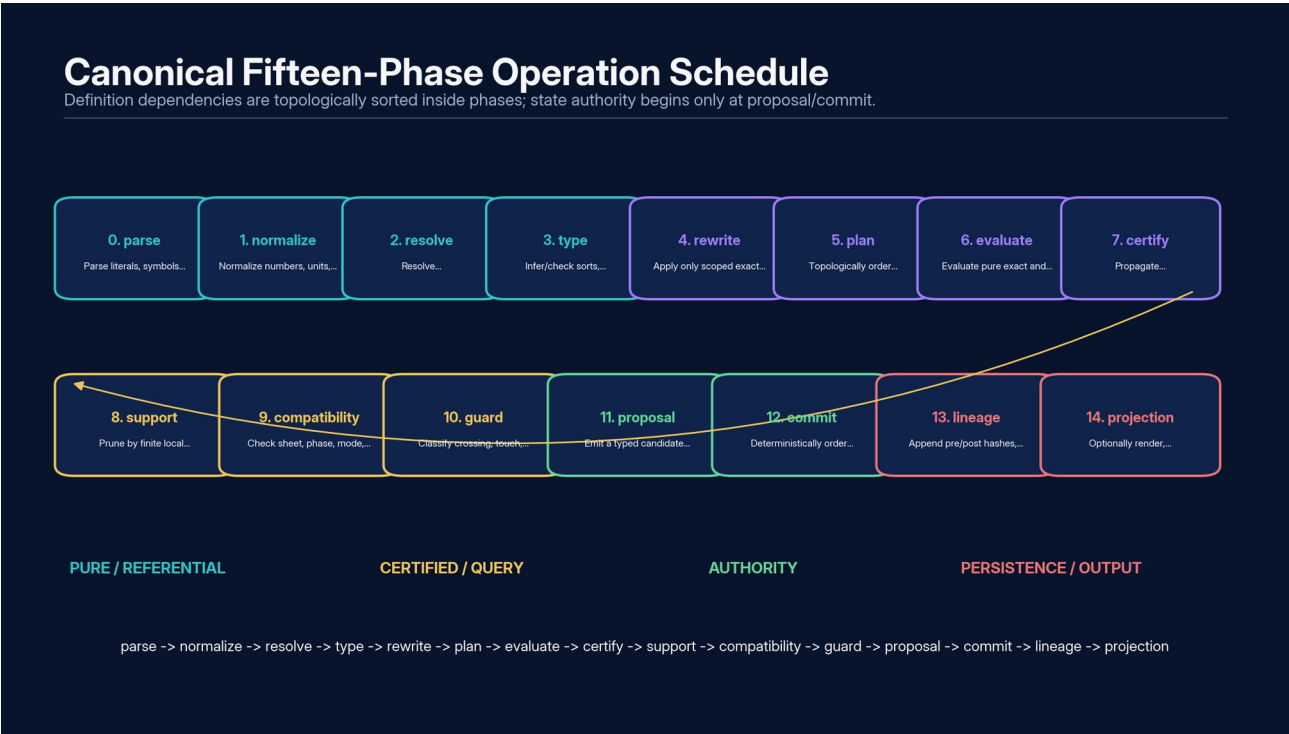


Figure 7. The canonical fifteen-phase schedule separates referential evaluation from query certification, authority and projection.

5.3 Dependency and cycle policy

Definitions are topologically sorted with stable priority and ID tie-breaking. An undeclared cycle is rejected. A declared feedback, recurrence or fixed-point operator must provide an initial state, monotonicity/contraction or other convergence evidence where applicable, iteration budget, termination status and bounded failure result.

6. Algebraic, Discrete and Functional Operator Families

The first six content domains provide the reusable operator grammar for notation, arithmetic, logic, order theory, finite sets/relations/combinatorics and sequence/functional aggregation. They deliberately separate exact discrete laws from profile-dependent numerical behavior.

Range	Domain	Representative families
M630-M645	Schema and typing	OperatorSpec, type universe, units, shapes, exactness, effects, hashes and extensions.
M646-M661	Parse/evaluation order	Precedence, fixity, grouping, dependency DAG, phases, reductions and composition.
M662-M677	Arithmetic/elementary	Constants, arithmetic, quotient/modulo, roots, rounding, logs, trig, gamma and modular families.
M678-M693	Logic/order theory	Boolean operators, comparisons, quantifiers, partial orders, lexicographic orders, infimum/supremum.
M694-M709	Sets/relations/combinatorics	Union/intersection, products, closures, quotients, cardinality, permutations and generating functions.
M710-M725	Sequences/functionals	Index, map/filter/fold/scan, stable sort/group, reductions, recurrence, composition and fixed points.

6.1 Reduction identity and empty cases

A variadic reduction records whether the empty collection has an identity, returns an option/failure state or is forbidden. Sum and product may use zero and one; minimum and maximum require a declared bounded lattice identity or a non-empty input. A general fold stores direction and initial accumulator.

6.2 Order theory as a substrate tool

Partial orders and topological orders are represented directly rather than forced into scalar sorting. This supports dependency evaluation, event precedence, interval containment, refinement and lattice-based fixed-point methods without pretending that every pair is comparable.

7. Linear, Geometric, Differential and Transform Families

M726-M789 cover vectors, matrices, tensors, complex/quaternion/geometric algebra, differential calculus and integration/measure/transforms. These domains are where composition direction, coordinate charts, rank, orientation and derivative availability most often change semantics.

Range	Domain	Key boundaries
M726-M741	Vectors, matrices and tensors	Dimension/shape checks; conjugating inner products; noncommutative matrix products; rank and solver status.
M742-M757	Complex, quaternion and geometric algebra	Conjugations are distinct; Hamilton/geometric products are ordered; versor sandwich action preserves typed grades.
M758-M773	Differential calculus	Limits, derivative/gradients/Jacobian/Hessian, exterior/Lie/covariant derivatives and AD with domain regularity.
M774-M789	Integration, measure and transforms	Domain, orientation, measure/Jacobian, singularities, quadrature error and transform convention are explicit.

7.1 Coordinate and topology boundary

A coordinate component, chart derivative or projected self-intersection is not automatically a physical vector, force or collision. Manifold transitions, orientation transport and quotient/gluing maps remain separate from dynamics and material claims. Integral theorem certificates require regularity, boundary and orientation hypotheses.

7.2 Transform conventions

```
function composition: (f o g)(x) = f(g(x))
column-vector transform chain: x_world = T_parent T_local x
quaternion rotation: v' = q v q^-1
Fourier/Laplace/wavelet transforms: normalization and sign convention are profile data
```

8. Probability, Optimization and Numerical Families

M790-M821 introduce probability/statistics/information and optimization/numerical/constraint families. They require explicit measures, seeds, confidence semantics, convergence tests, residuals and budgets. A stochastic result without its distribution and seed/novelty record is not replayable.

Range	Family	Reference coverage	Specified boundary
M790-M805	Probability/statistics/information	Mean, variance, covariance, entropy, KL, mutual information and seeded samples.	General distributions, Bayesian systems and hypothesis workflows require domain libraries and audit.
M806-M821	Optimization/numerical/constraints	Gradient descent, bisection, interpolation/least-squares contracts, interval certification.	Full LP/QP/KKT/Krylov/PDE solvers are typed contracts, not dependency-free production solvers.

8.1 Convergence and failure status

Every iterative method returns a result plus residual or gradient norm, iteration count and status such as converged, endpoint, infeasible, singular, unbounded, ambiguous or budget_exhausted. A plausible number without solver status is not a certified substrate result.

8.2 Probability is not authority

Confidence, posterior probability or classifier score can contribute to a guard, but it does not bypass support, compatibility, provenance, numeric error or application policy. Exogenous randomness must be seeded for reproducibility or logged as irreducible external novelty.

9. Symbolic Rewrite, Proof and Metaprogramming

M854-M869 add symbols, binding, substitution, alpha/beta operations, simplification, canonicalization, pattern matching, rewrite rules, term ordering, common-subexpression elimination, solver registries, assumptions, proof obligations, bounded grammar and operator generators.

9.1 Rewrite rule record

```
RewriteRule = (id, pattern, replacement, side_conditions,
               orientation, term_order, termination_budget,
               equality_or_approximation_status, provenance, hash)
```

An equality-preserving rewrite is distinct from an approximation, compiler lowering or heuristic simplification. Side conditions must be checked before the rule is used. Term ordering and budgets prevent an uncontrolled rewrite loop. Common-subexpression elimination is safe only for pure nodes or explicitly shareable effects.

9.2 Proof boundary

PROOF AND THEOREM STATUS

The catalog can carry assumptions, obligations and certificates, but the dependency-free runtime is not a complete theorem prover. A proof certificate is meaningful only under the declared logic, axiom set, checker and trusted computing base.

9.3 Operator generators

The operator-generator and open-namespace mechanisms are the structural answer to the request for “all possible” operators. A domain can create parameterized families - for example, p-norms, tensor contractions, special functions, differential operators or graph semirings - while inheriting the same type, order, exactness, effect, budget, evidence and UGTS-handoff contracts.

10. Exactness, Interval and Certification Discipline

10.1 Status classes

Class	Meaning	Promotion rule
Exact symbolic	Algebraic expression retained without numerical rounding.	Use only under validated symbolic laws and domains.
Exact discrete	Integer, bit-vector, finite-set or combinatorial result.	Overflow/range profile must be explicit.
Floating approximate	Binary floating result with platform/profile limits.	Record precision, reduction order and tolerance.
Interval/residual certified	True value is enclosed or residual bounded.	Guard may commit only if the entire certificate classifies the event.
Stochastic/estimated	Distribution or estimator with seed/data assumptions.	Record uncertainty and external novelty.
Bounded iterative	Algorithm stopped under convergence or work budget.	Return status; never disguise budget exhaustion as exact.
Presentation-only	Projection for viewing/export.	Cannot mutate authoritative state.
Specified contract	Typed but not implemented by reference runtime.	Route to a suitable audited domain adapter.

10.2 Interval guard

```
authoritative relation: f in [lower, upper]
negative   when upper < -margin
positive   when lower > +margin
exact_zero when lower = upper = 0
otherwise  ambiguous_guard
```

The guard must not promote a result when the error interval straddles the decision boundary. Improving the measurement, algebra, precision or sample budget is preferable to silently choosing a side.

10.3 Packing and quantization

The SCLP and GPU-native sources already establish the relevant boundary: width reduction is not correctness or semantic equivalence. Version 4.2 carries quantization range, step, rounding, saturation, decode function and propagated error in the operator profile. A packed value may serve as an index or cache key without becoming complete state.

11. Determinism, Parallel Reduction, Compression and Exchange

11.1 Deterministic joins

Pure independent nodes may run in parallel, but the join order is a profile. Exact integer/bit operations may use any lawfully equivalent tree; floating, interval, noncommutative and stateful operations preserve or explicitly certify the chosen order. Stable sorted-pairwise reduction is supplied as a reproducible option, not a universal accuracy optimum.

11.2 Canonical exchange

Record	Canonical rule	Boundary
JSON value	sorted keys, compact separators, integral-float normalization, tagged tuples/sets/complex/fractions	integrity format; not a signature
Expression	profile ID + explicit canonical AST	whitespace/token spelling does not define meaning
Definition	content hash excludes the hash field	hash change signals semantic record change
Event	monotonic sequence + pre/post hashes + sorted patch	digest detects divergence; full state/lineage remains authoritative
Seed reconstruction	seed + versioned algorithm + parameters + external deltas	valid only when deterministic reconstruction is demonstrated

11.3 Compression and budget records

Cost and memory records distinguish nominal width, retained semantics, decode cost, event density, checkpoint density and reconstruction failure. A codebook or opcode can select a finite family; it cannot encode an arbitrary object without a versioned external definition. A seed is not encryption and does not prove provenance.

12. UGTS Event Authority, Replay and Lineage

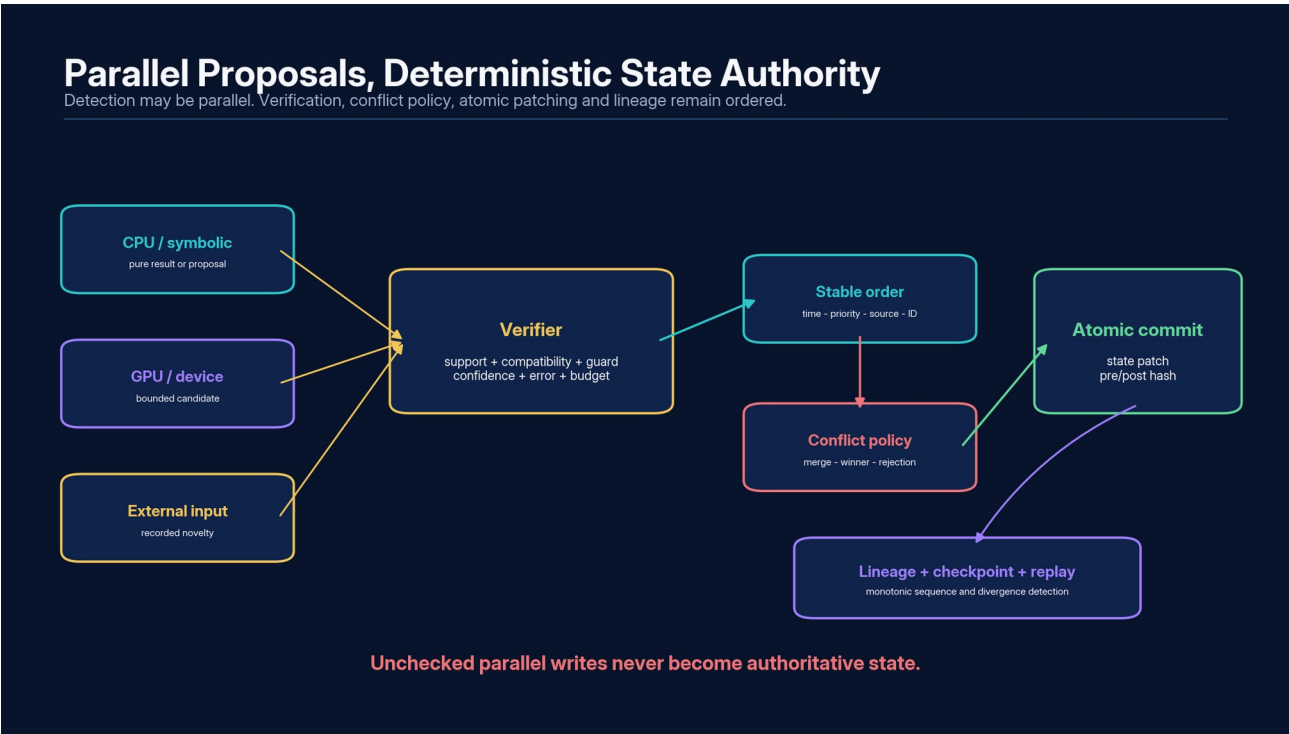


Figure 8. Parallel operator results become proposals; verification and commit remain deterministic state authority.

12.1 Verified proposal predicate

```
verified = support_ok
          and compatibility_ok
          and guard_status in accepted_statuses
          and confidence >= confidence_floor
          and numeric_error <= event_margin
          and work_budget_status is acceptable
```

12.2 Simultaneous events

The default total order is event time, descending priority, source and proposal ID. Proposals modifying the same target field at the same event time conflict. A versioned application may replace winner-takes-first with a proven commutative merge, but the merge law and evidence must be explicit. Rejections are typed results, not silent losses.

12.3 Replay

Each committed event stores a monotonic sequence, patch, pre-state hash, post-state hash and lineage label. Replay starts from the initial state or checkpoint, verifies the pre-hash, applies the patch, verifies the post-hash and stops at the first divergence. Hashes are checksums for consistency, not complete identity or legal evidence.

13. Reference Implementation and CLI

13.1 Package modules

Module	Responsibility
canonical.py	Canonical JSON-compatible values, SHA-256 content addresses and record verification.
catalog.py	Load, search, summarize and validate the M630-M885 operator catalog.
order.py	Load order profiles and exchange canonical expression ASTs.
evaluator.py	Restricted safe expression evaluator and deterministic trace.
interval.py	Finite interval arithmetic and guard classification.
linalg.py	Vectors, matrix products, determinant, solve, rank and Kronecker product.
numeric.py	Finite differences, integration, root finding, optimization, statistics and information measures.
graph.py	Deterministic graph normalization, reachability, shortest path and topological sort.
events.py	Proposal verification, stable ordering, atomic patching, lineage and replay.
validation.py / demo.py / cli.py	Project/catalog/profile validation, integrated demonstration and command-line interface.

13.2 CLI workflow

```
PYTHONPATH=src python -m ugts_kc42 info
PYTHONPATH=src python -m ugts_kc42 catalog --search gradient --json
PYTHONPATH=src python -m ugts_kc42 eval "2 + 3 * 4 ** 2" --trace
PYTHONPATH=src python -m ugts_kc42 validate examples/general_operator_order_project.json
PYTHONPATH=src python -m ugts_kc42 demo examples/demo_output.json
```

13.3 Safety boundary

The evaluator accepts an expression-only AST subset with a fixed operator/function registry. It rejects imports, attribute access, statements, lambdas, comprehensions and arbitrary named calls. The reference code is an inspectable oracle and educational substrate, not a production symbolic/numerical platform.

14. Deterministic Demonstration

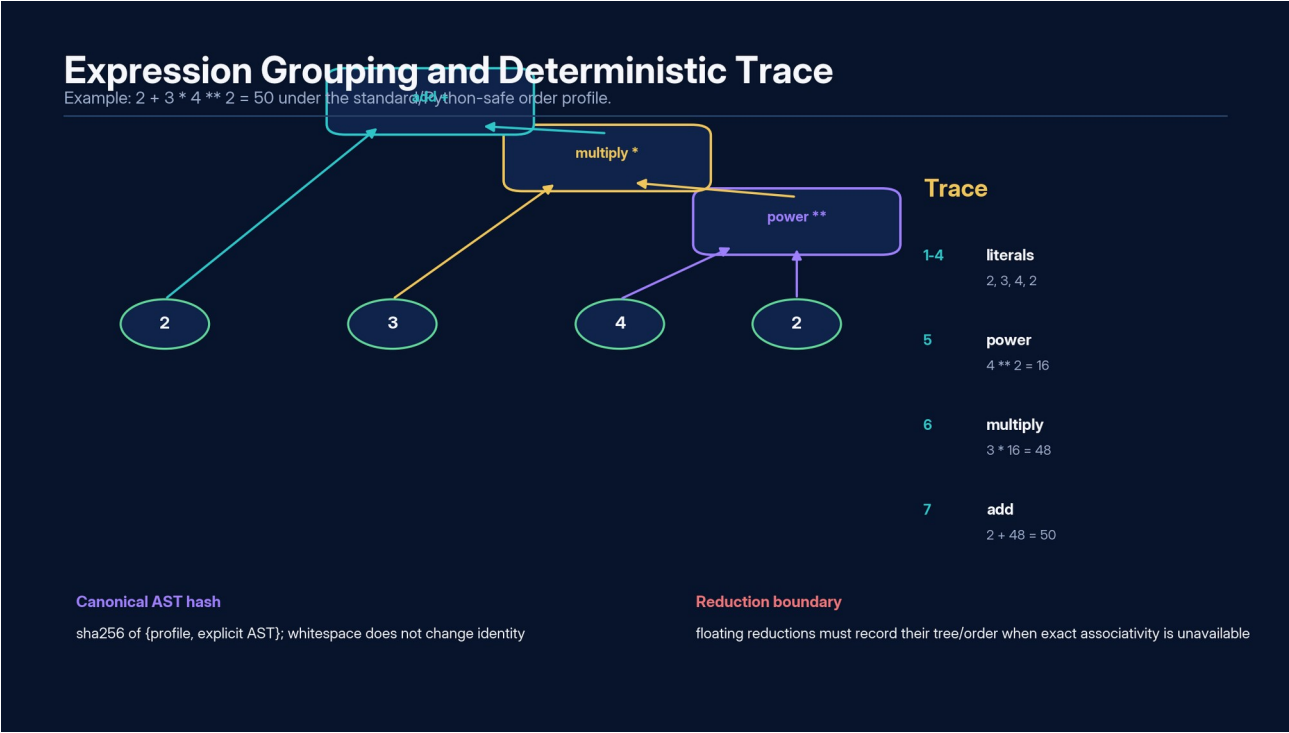


Figure 9. The expression tree and trace make precedence and evaluation order inspectable.

14.1 Integrated scenario

The demonstration evaluates a mixed scalar/vector expression, propagates an interval guard, solves a linear system, estimates a derivative and integral, brackets a root, computes entropy and mutual information, performs graph reachability/shortest path, topologically orders definitions, verifies competing proposals, commits a deterministic winner and confirms replay hash equality.

Metric	Captured result
Expression	clamp(dot([1, 2, 3], [4, 5, 6]) / 8 + sin(pi / 2) ** 2, 0, 10)
Expression result	5.0
Interval class	ambiguous_guard
Matrix determinant	10
sqrt(2) root	1.414213562373
Graph shortest path	{'cost': 3.0, 'path': ['A', 'B', 'C', 'D'], 'status': 'ok'}
Accepted proposal IDs	p-high, p-mode
Rejected proposals	[{'conflicts': [{'path': 'world.value', 'winner': 'p-high'}], 'proposal_id': 'p-low', 'reasons': ['patch_conflict']}, {'proposal_id': 'p-bad', 'reasons': ['outside_support']}]
Final/replay hash match	True
Demo content hash	sha256:abc476fd1121a63ceed9d489097c4864b6f64ac9fdce1cfac2d264fab4908fe

14.2 Interpretation

EVIDENCE BOUNDARY

The demonstration proves deterministic internal behavior of the bundled reference subset. It does not prove that every cataloged advanced operator is numerically implemented, that all mathematical problems terminate, or that another platform will be bit-identical outside the selected profile.

15. Validation and Release Evidence

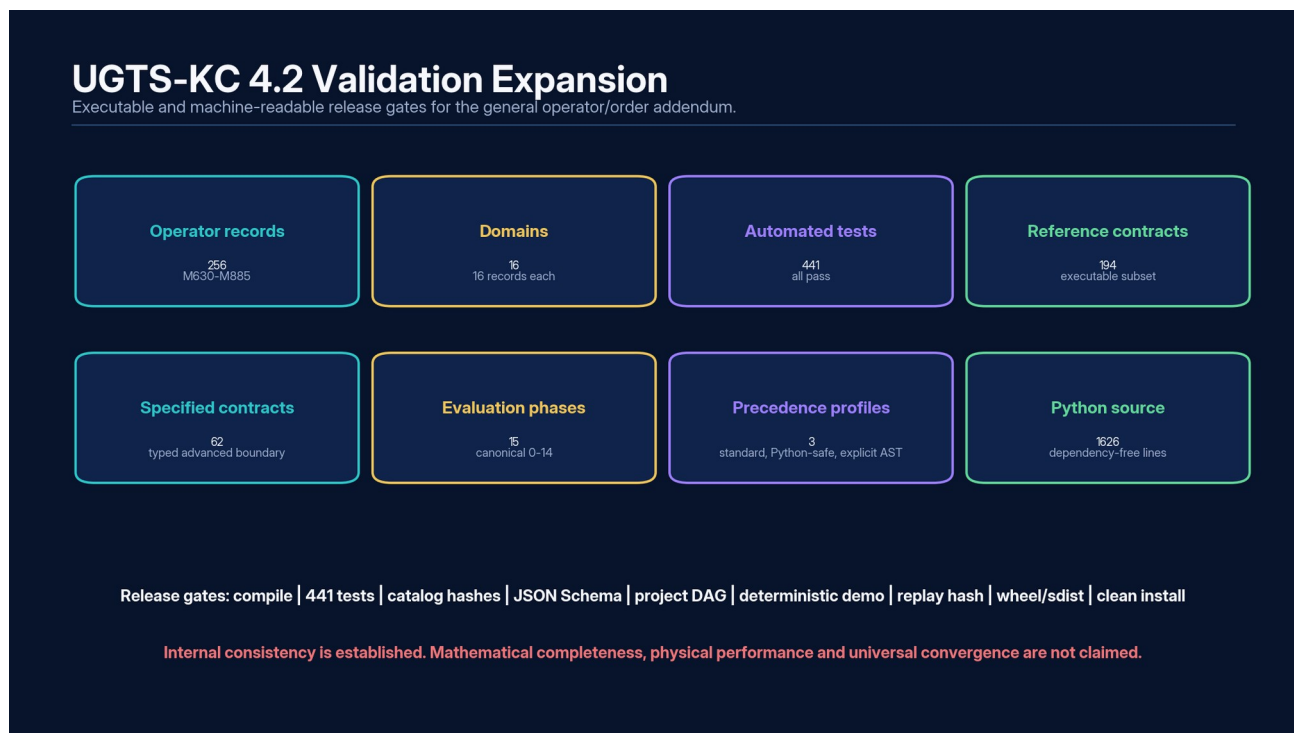


Figure 10. Release gates and exact measured package metrics.

Gate	Result	What is established
Python compile	PASS	All reference modules compile.
Automated tests	441/441 PASS	Every catalog row plus canonicalization, expressions, intervals, numerics, graphs, events, schema, demo and CLI.
Catalog	PASS	M630-M885 contiguous; 256 hashes valid; 16 domains of 16.
JSON Schema/project	PASS	Draft 2020-12 example structurally valid; 12 definition hashes and DAG valid.
Order profiles	PASS	Three unique profiles; phase indices contiguous 0-14.
Demo/replay	PASS	Repeat content hash and final/replay state hash match.
Distributions	PASS	Wheel and source archive built.
Clean install	PASS	Wheel installed in an isolated environment and CLI info executed.
PDF/DOCX preflight	PASS	35-page final render opens, is tagged and text-bearing; all pages were visually inspected and PDF preflight recorded no blocking issue.

15.1 What the tests do not establish

They do not establish universal mathematical completeness, theorem soundness for arbitrary external proof systems, production cryptographic safety, physical-device performance, cross-vendor floating-point identity, or suitability for a safety-critical application. Those require application-specific libraries, measurements, review and promotion gates.

16. Compatibility, Migration and Catalog Continuity

16.1 Additive integration

Existing UGTS modules and schemas remain valid. A profile can adopt only the general operator record and explicit order data without replacing its domain runtime. Earlier mechanism IDs and semantics are not rewritten. The general layer may reference a domain operator by stable ID while keeping domain-specific source definitions and native implementations in their existing packages.

16.2 Migration steps

1. Assign a stable operator ID and semantic version.

2. Declare arity, domain, codomain, units, shape, exactness and effects.
3. Select or define a precedence/fixity profile; prefer explicit AST for exchange.
4. List dependencies and validate a deterministic topological order.
5. Record composition and reduction order where laws do not make it irrelevant.
6. Separate pure evaluation from proposal, commit and projection.
7. Attach residual, interval, uncertainty, work and provenance status.
8. Preserve pre/post hashes, checkpoint and lineage for state changes.

16.3 Fallback policy

When a target domain already has a mature audited implementation, the 4.2 reference runtime should act as schema, oracle and event-handoff layer rather than reimplementing the domain. The kill criterion is explicit: use the conventional implementation when it is simpler, faster, safer or more accurate at equal error.

17. Boundaries and Kill Criteria

17.1 Non-claims

- No finite list is “all mathematics”; open-ended coverage comes from the general record and operator generators.
- No complete theorem prover, computer algebra system, PDE solver, optimizer or special-function library is claimed.
- No universal convergence, termination, stability, exactness, $O(1)$, compression or hardware-throughput result is claimed.
- No content hash proves mathematical truth, identity, ownership, authorization or chain of custody.
- No host test implies phone, GPU, thermal, energy or safety performance.

17.2 Numerical and semantic kill criteria

- Type, unit, shape, coordinate or domain conventions are ambiguous.
- A rewrite uses a law outside its scope or changes semantics without an approximation label.
- Error, quantization or interval width exceeds the event margin or changes event ordering.
- A solver lacks a bounded failure state or exceeds its declared work budget.
- Reduction/parallel order changes the result beyond tolerance and is not recorded.
- A packed/seed record cannot reconstruct the required state, uncertainty and lineage.

17.3 Order and authority kill criteria

- A dependency cycle is undeclared or lacks fixed-point initialization/convergence semantics.
- Effectful operations are reordered as if pure.
- Simultaneous patches conflict without priority, commutative merge or conflict event.
- Replay fails pre/post hash checks or external novelty is unrecorded.
- Projection or model output is treated as authoritative without proposal verification.

PRAGMATIC RULE

Simplify or reject the general layer when it adds more schema, certification and event density than the avoided ambiguity or when a conventional domain system is better at the same correctness/error target.

18. Final Upgraded Definition

FORMAL DEFINITION

UGTS-KC 4.2 is a finite, content-addressed and extensible mathematical operator substrate in which operator meaning, types, syntax, precedence, associativity, dependencies, composition, phase, reduction, parallel, event and replay order are explicit data. Pure results remain separate from authority. Any state change returns to support, compatibility, guard, verified proposal, deterministic commit, transition and lineage.

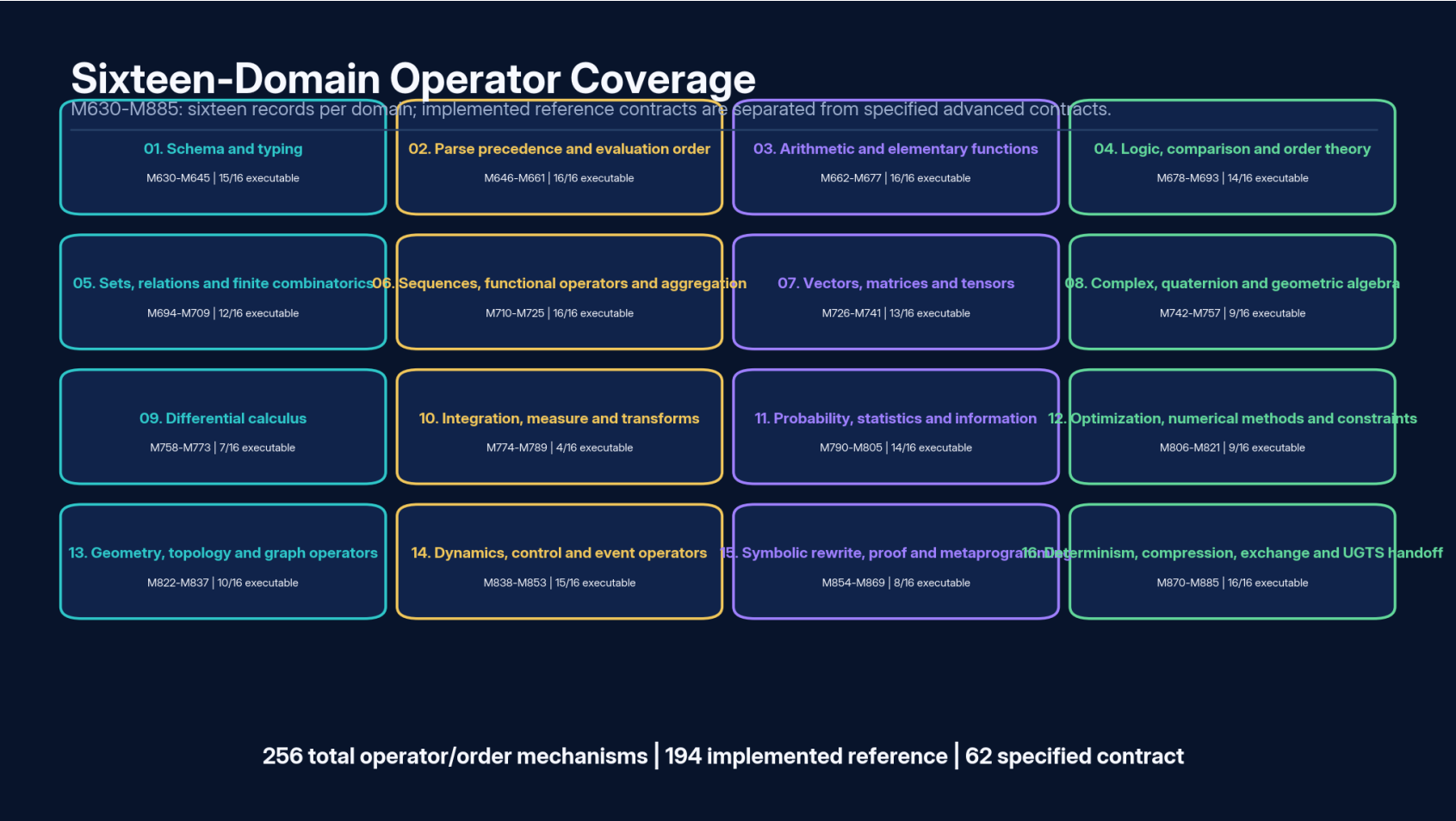
The strongest upgrade is not the number of catalog rows. It is that ambiguity is now representable and rejectable. A symbol cannot silently choose its own domain, a transform cannot silently commute, a reduction cannot silently change trees, a probabilistic score cannot silently become state, and a projection cannot silently become truth.

UGTS-KC 4.2 = retained UGTS + OperatorSpec + OrderProfile + Certificate

complete release claim: all 256 shipped records, hashes, profiles, phases, tests and artifacts are present
not claimed: closure of mathematics, universal solvers, universal proof or physical performance

Appendix A. Complete Compact Catalog M630-M885

The machine-readable CSV/JSON files contain full definitions, integration notes, provenance and content hashes. The following tables are the complete human-readable compact index. Each of the sixteen domains contains exactly sixteen records.



Appendix overview. Sixteen domains provide the finite release index; open operator generators provide future extension without a false closure claim.

CATALOG READING RULE

EXEC identifies an executable bounded reference mechanism or directly checked record contract. SPEC identifies a typed advanced contract whose full domain implementation is not claimed by the dependency-free runtime.

A.1 Schema and typing (M630-M645)

ID	Operator / mechanism	Typed signature	State	Normalized definition
M630	General operator record	OperatorSpec -> validated operator	EXEC	A mathematical operator is a content-addressed record carrying stable ID, arity, typed domain, codomain, parameters, laws, exactness, order contract, provenance and capabilities.
M631	Type and sort universe	TypeExpr -> normalized sort	EXEC	Declares scalar, numeric tower, Boolean, bit-vector, interval, sequence, set, relation, function, vector, matrix, tensor, graph, manifold, distribution, state and event sorts with explicit parameters.
M632	Arity and variadic contract	arity x operands -> admission	EXEC	Records nullary, unary, binary, ternary or variadic arity, minimum and maximum operand counts, and whether empty reduction has an identity.
M633	Domain and codomain signature	(A1,...,An) -> B	EXEC	Every operand position and result has a declared type expression; polymorphic variables and constraints are explicit rather than inferred from prose.
M634	Unit and dimension signature	units(Ai) -> units(B)	SPEC	Physical or semantic dimensions are attached to operands and outputs; addition requires compatible dimensions and multiplication composes dimension exponents.
M635	Shape, rank and broadcasting signature	shape(Ai) -> shape(B)	EXEC	Array-like operands declare shape variables, rank, axis meaning, broadcast rules and output shape; silent shape collapse is rejected.
M636	Total, partial and multivalued status	input -> value bounded failure	EXEC	An operator declares whether it is total, partial, set-valued or relation-valued, plus domain guards and failure reason codes.
M637	Purity and effect class	operator -> effect class	EXEC	Separates pure mathematical evaluation from state read, proposal emission, authoritative mutation, I/O and downstream projection effects.
M638	Exactness and error model	operator x profile -> error semantics	EXEC	Marks exact symbolic, exact discrete, floating, interval-bounded, stochastic, approximate or presentation-only semantics and carries an error contract.
M639	Capability flags	operator -> capability set	EXEC	Declares available derivative, gradient, Jacobian, Hessian, adjoint, inverse, monotonicity, convexity, vectorization and serialization capabilities.
M640	Algebraic law declaration	operator -> scoped laws	EXEC	Associativity, commutativity, identity, inverse, idempotence, distributivity, monotonicity and homomorphism laws are data and may be used only when their scope is satisfied.
M641	Precondition, invariant and guard contract	input x state -> guard status	EXEC	Domain restrictions, numerical floors, topology invariants and event margins are first-class predicates evaluated before promotion of a result.
M642	Content-addressed operator identity	canonical operator record -> digest	EXEC	Canonical JSON excluding the hash field is SHA-256 hashed; the digest verifies unchanged semantics but does not prove legal authorship or mathematical truth.
M643	Explicit dependency edge	operator graph -> DAG or rejection	EXEC	Composite operators list stable IDs of definitions they depend on; missing references, duplicate IDs and undeclared cycles are validation failures.
M644	Version and migration record	old schema x migration -> new schema	EXEC	A semantic change creates a new version and explicit migration; aliases cannot silently change precedence, units, domains, laws or numerical meaning.
M645	Open extension namespace	namespace x operator family -> extension	EXEC	Profiles may add operators under collision-resistant namespaces while retaining the general record, order, evidence and UGTS handoff contracts.

EXEC = executable bounded reference mechanism or directly checked record contract. SPEC = typed advanced contract; full domain implementation is not claimed.

A.2 Parse precedence and evaluation order (M646-M661)

ID	Operator / mechanism	Typed signature	State	Normalized definition
M646	Token precedence table	token x profile -> precedence	EXEC	Maps every symbolic or keyword operator to a numeric binding power in a named parser profile; ambiguous symbols require an explicit profile.
M647	Associativity contract	operator -> associativity	EXEC	Declares left, right, non-associative or variadic grouping; exponentiation is right-associative in the standard mathematical profile.
M648	Fixity and mixfix contract	operator -> syntactic form	EXEC	Declares prefix, infix, postfix, function-call, indexing, delimiter or mixfix syntax independently from semantic arity.
M649	Parenthesized grouping authority	tokens -> grouped AST	EXEC	Explicit grouping overrides profile precedence without changing the underlying operator definitions; the resulting AST is authoritative.
M650	Function application precedence	f(args) -> application node	EXEC	Function calls, indexing and explicit operator application bind more tightly than ordinary infix arithmetic unless a profile declares otherwise.
M651	Exponent and root grouping	base x exponent -> power AST	EXEC	Power chains group according to the selected profile and unary signs are distinguished from signed literals; roots are explicit operators, not ambiguous fractional exponents.
M652	Unary sign versus power rule	tokens -> unambiguous unary AST	EXEC	The standard profile parses $-x^2$ as $-(x^2)$, while an alternate language profile may differ; the selected rule is stored in the trace.
M653	Multiplicative and additive levels	tokens -> arithmetic AST	EXEC	Products, quotients and remainders bind more tightly than sums and differences; implicit multiplication is disabled by default to avoid symbol collisions.
M654	Comparison chain order	comparison chain -> predicate	EXEC	A chain $a < b \leq c$ is represented as a conjunction sharing the middle operand and is not rewritten as nested Boolean-to-number comparison.
M655	Boolean precedence order	Boolean tokens -> predicate AST	EXEC	The standard profile orders not above and above xor above or above implication; three-valued profiles preserve unknown propagation.
M656	Conditional and case order	selector x branches -> selected value	EXEC	Piecewise, if-then-else and match/case constructs evaluate the selector before exactly one selected branch unless a symbolic profile requests branch retention.
M657	Dependency topological order	definition DAG -> ordered IDs	EXEC	Definitions are evaluated only after their declared dependencies; deterministic tie-breaking uses phase, priority and stable ID.
M658	Normative phase schedule	program -> phase trace	EXEC	The general evaluator uses parse, normalize, resolve, type, rewrite, plan, evaluate, certify, support, compatibility, guard, proposal, commit, lineage and projection phases.
M659	Stable equal-rank tie break	ready set -> deterministic next node	EXEC	When multiple independent nodes are eligible, the profile orders them by explicit priority then stable content ID, never by container hash or thread timing.
M660	Reduction tree order	sequence x reduction profile -> result	EXEC	A fold records left, right, pairwise, balanced, Kahan-like, exact or implementation-defined reduction order; floating reductions are never reordered silently.
M661	Composition order and trace	operator sequence x input -> trace	EXEC	For $f \circ g$ the substrate records that g is applied first; noncommuting transformations preserve serialized order and intermediate values in the trace.

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A.3 Arithmetic and elementary functions (M662-M677)

ID	Operator / mechanism	Typed signature	State	Normalized definition
M662	Literal and named constant	$\{\} \rightarrow \text{scalar}$	EXEC	Nullary exact or profiled constants including integers, rationals, pi, e and declared application constants with explicit precision.
M663	Unary plus and negation	$A \rightarrow A$	EXEC	Identity and additive inverse over additive structures; negation is distinct from bitwise complement and Boolean not.
M664	Addition	$A \times A \rightarrow A$	EXEC	Binary or variadic addition over a declared additive structure with optional identity zero and no floating reassociation unless permitted.
M665	Subtraction	$A \times A \rightarrow A$	EXEC	Addition of an additive inverse where available; set difference and geometric subtraction remain separate operator IDs.
M666	Multiplication	$A \times A \rightarrow A$	EXEC	Product under a declared algebra, including scalar, elementwise or ring multiplication; matrix and geometric products use distinct records.
M667	Division	$A \times A_{\text{nonzero}} \rightarrow B$	EXEC	Multiplication by an inverse on the guarded nonzero domain; quotient type and zero policy are explicit.
M668	Quotient, modulo and remainder family	$\text{integer} \times \text{integer}_{\text{nonzero}} \rightarrow \text{integer pair}$	EXEC	Distinguishes Euclidean quotient/modulo, truncating division/remainder and language-specific sign conventions.
M669	Power and root family	$\text{base} \times \text{exponent} \rightarrow \text{value}$	EXEC	Integer, rational, real or complex exponentiation and explicit nth-root operators with branch and domain policy.
M670	Absolute value, magnitude and sign	$A \rightarrow \text{ordered scalar or unit}$	EXEC	Returns an ordered magnitude/norm or sign-like unit under the declared scalar or algebraic type; zero sign policy is explicit.
M671	Floor, ceiling, truncation and rounding	$\text{real} \times \text{policy} \rightarrow \text{discrete value}$	EXEC	Quantization family with explicit tie policy, radix, precision and signed-zero behavior.
M672	Minimum, maximum and clamp	$\text{ordered values} \rightarrow \text{selected value}$	EXEC	Order-dependent selection with total/partial order policy, stable ties and interval-aware variants.
M673	Exponential and logarithm family	$\text{scalar} \rightarrow \text{scalar}$	EXEC	exp, log and log-base operators with branch/domain policy and exact symbolic capability metadata.
M674	Trigonometric family	$\text{angle} \rightarrow \text{scalar}$	EXEC	sin, cos, tan and reciprocal circular functions with angle-unit normalization and singularity guards.
M675	Inverse and hyperbolic trigonometric family	$\text{scalar} \rightarrow \text{angle or scalar}$	EXEC	Inverse circular and hyperbolic functions with explicit principal branch and codomain interval.
M676	Factorial, gamma and binomial family	$\text{integer or scalar} \rightarrow \text{scalar}$	EXEC	Discrete factorial and combinations plus gamma extension under declared domains; overflow and approximation status are visible.
M677	GCD, LCM and modular arithmetic family	$\text{integers} \rightarrow \text{integer}$	EXEC	Number-theoretic operators including gcd, lcm, modular reduction, modular inverse and modular exponentiation with exact integer semantics.

EXEC = executable bounded reference mechanism or directly checked record contract. SPEC = typed advanced contract; full domain implementation is not claimed.

A.4 Logic, comparison and order theory (M678-M693)

ID	Operator / mechanism	Typed signature	State	Normalized definition
M678	Boolean and truth literal	$\{\} \rightarrow \text{truth value}$	EXEC	Nullary truth values under two-, three- or four-valued logic profiles; unknown and inconsistent are not silently coerced to false.
M679	Logical negation	$\text{Truth} \rightarrow \text{Truth}$	EXEC	Truth negation under the selected logic algebra, distinct from arithmetic sign and bit complement.
M680	Conjunction	$\text{Truth}^n \rightarrow \text{Truth}$	EXEC	Logical and with declared short-circuit or symbolic evaluation and truth-table profile.
M681	Disjunction and exclusive disjunction	$\text{Truth}^n \rightarrow \text{Truth}$	EXEC	Logical or and xor families with explicit n-ary parity semantics and short-circuit status.
M682	Implication and equivalence	$\text{Truth} \times \text{Truth} \rightarrow \text{Truth}$	EXEC	Material or profile-defined implication, biconditional and logical equivalence; proof implication remains separately typed.
M683	Exact equality and inequality	$A \times A \rightarrow \text{Truth}$	EXEC	Structural or mathematical equality under a declared type; hash equality is never substituted for semantic equality without a proof contract.
M684	Ordered comparison	$A \times A \rightarrow \text{order status}$	EXEC	Less/greater and non-strict comparisons under a declared preorder, partial order or total order; incomparable is a possible result.
M685	Approximate and interval equality	$\text{numeric} \times \text{numeric} \times \text{tolerance} \rightarrow \text{status}$	EXEC	Absolute, relative, ULP or interval-overlap comparison with explicit tolerance and transitivity warning.
M686	Membership and containment	$\text{element} \times \text{container} \rightarrow \text{Truth}$	EXEC	Element-of, subset, proper subset and containment predicates across sets, intervals, spaces and supports.
M687	Predicate and characteristic operator	$A \rightarrow \text{Truth or indicator}$	EXEC	Maps values to truth/status and optionally to an indicator in a declared codomain; predicate identity is retained.
M688	Universal and existential quantifier	$\text{domain} \times \text{predicate} \rightarrow \text{Truth/status}$	SPEC	Finite, symbolic or measure-qualified forall/exists operators with explicit domain and bounded enumeration policy.
M689	Preorder, partial order and total order record	$\text{relation} \rightarrow \text{order certificate}$	EXEC	Stores relation properties such as reflexive, antisymmetric, transitive and total, with executable finite validators when applicable.
M690	Strict and non-strict order conversion	$\text{order relation} \rightarrow \text{paired relation}$	SPEC	Converts between $<$ and $\leq$ forms only when equality and order laws justify it; incomplete orders preserve incomparability.
M691	Lexicographic, product and shortlex order	$\text{component orders} \rightarrow \text{composite order}$	EXEC	Constructs deterministic composite orders from component orders, including length-first and coordinatewise variants.
M692	Well-foundedness and topological order	$\text{relation or DAG} \rightarrow \text{certificate/order}$	EXEC	Records or checks absence of descending chains in bounded structures and provides deterministic topological ordering for finite DAGs.
M693	Infimum, supremum and order statistics	$\text{ordered collection} \rightarrow \text{bound/statistic}$	EXEC	Computes lower/upper bounds, minima/maxima, ranks and quantiles under a declared order and existence policy.

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A.5 Sets, relations and finite combinatorics (M694-M709)

ID	Operator / mechanism	Typed signature	State	Normalized definition
M694	Set, multiset and finite-map literal	elements -> collection	EXEC	Constructs unordered sets, counted multisets or key-value maps with canonical element/key encoding and duplicate policy.
M695	Union	Collection x Collection -> Collection	EXEC	Set or multiset union with declared multiplicity rule; geometric CSG union is a separate field operator.
M696	Intersection	Collection x Collection -> Collection	EXEC	Common membership under set or multiset semantics with deterministic representation.
M697	Difference and complement	Collection x Collection/Universe -> Collection	EXEC	Relative difference or complement against an explicit universe; no implicit universal set is assumed.
M698	Symmetric difference	Collection x Collection -> Collection	EXEC	Elements present in exactly one operand under set semantics, or a declared multiset parity/count rule.
M699	Cartesian product	$A \times B \rightarrow \text{Set}[(A,B)]$	EXEC	Ordered tuple construction over factor collections with finite budget and product-order metadata.
M700	Power set and finite subset family	finite set -> subset family	EXEC	Enumerates or symbolically denotes subsets with explicit cardinality budget; unbounded materialization is rejected.
M701	Relation composition	$R:A \leftrightarrow B \times S:B \leftrightarrow C \rightarrow A \leftrightarrow C$	EXEC	Composes typed binary relations by existentially matching the intermediate sort; ordering of composition is explicit.
M702	Inverse and converse relation	$R:A \leftrightarrow B \rightarrow R^{-1}:B \leftrightarrow A$	EXEC	Swaps relation endpoints without claiming a functional inverse; multiplicity and provenance are retained.
M703	Reflexive, symmetric and transitive closure	relation -> closed relation	EXEC	Finite graph/relation closure operators with algorithm and budget; closure is not silently applied to authoritative data.
M704	Equivalence class and quotient set	set x equivalence -> quotient	SPEC	Forms classes only after an equivalence-relation certificate; representative selection and lineage are explicit.
M705	Cardinality and counting measure	collection -> cardinal	EXEC	Returns exact finite size, symbolic cardinality or bounded count status; bytes and semantic information are distinct metrics.
M706	Permutation and combination	$n \times k \times \text{policy} \rightarrow \text{count/family}$	EXEC	Exact nPk, nCk and iterator families with order, repetition and budget policy.
M707	Partition and composition family	object x constraints -> partition family	SPEC	Integer/set partitions and ordered compositions under bounded generation with canonical ordering.
M708	Inclusion-exclusion and Mobius inversion	finite family/function -> count/transformed function	SPEC	Finite combinatorial transforms over subset or poset incidence with explicit domain and exact arithmetic.
M709	Generating function and recurrence descriptor	sequence/recurrence -> formal series	SPEC	Represents ordinary/exponential generating functions and finite recurrences as typed definitions; convergence and coefficient extraction are separate capabilities.

EXEC = executable bounded reference mechanism or directly checked record contract. SPEC = typed advanced contract; full domain implementation is not claimed.

A.6 Sequences, functional operators and aggregation (M710-M725)

ID	Operator / mechanism	Typed signature	State	Normalized definition
M710	Sequence, tuple and stream record	elements/generator $\rightarrow$ sequence	EXEC	Ordered finite or lazily bounded values with element type, length status, seed and traversal order.
M711	Indexing and slicing	sequence $\times$ index/range $\rightarrow$ value/sequence	EXEC	Selects elements or ranges with explicit base, negative-index, bounds and half-open/closed interval policy.
M712	Concatenation and interleave	sequences $\rightarrow$ sequence	EXEC	Joins ordered sequences or interleaves them by a declared schedule without losing source boundaries.
M713	Map	$(A \rightarrow B) \times \text{Seq}[A] \rightarrow \text{Seq}[B]$	EXEC	Applies a pure operator elementwise while retaining element order and per-element failure status.
M714	Filter and partition	predicate $\times$ $\text{Seq}[A] \rightarrow \text{Seq}[A]$ or pair	EXEC	Retains or splits elements by a predicate; stable input order is the default.
M715	Fold and reduce	binary op $\times$ identity $\times$ $\text{Seq}[A] \rightarrow A$	EXEC	Combines a sequence with explicit identity, direction and reduction tree; algebraic laws may permit optimized trees only under profile.
M716	Scan and prefix accumulation	op $\times$ identity $\times$ $\text{Seq}[A] \rightarrow \text{Seq}[A]$	EXEC	Returns all intermediate fold states with inclusive/exclusive and direction policy.
M717	Zip, unzip and alignment	$\text{Seq}[A] \times \text{Seq}[B] \rightarrow \text{Seq}[(A,B)]$	EXEC	Pairs or groups sequences by position, key, time or interpolation policy; truncation/padding is explicit.
M718	Stable sort	$\text{Seq}[A] \times \text{order} \rightarrow \text{Seq}[A]$	EXEC	Orders values by a declared total/key order with stable equal-key behavior; partial-order sorting returns a linear-extension status.
M719	Unique, group-by and histogram	$\text{Seq}[A] \times \text{key} \rightarrow \text{groups}$	EXEC	Deduplicates or groups by exact/canonical key and produces counts or buckets with stable first-occurrence ordering.
M720	Sum and product reduction	$\text{Seq}[\text{number}] \rightarrow \text{number}$	EXEC	Specialized reductions with exact, pairwise, compensated or interval accumulation profiles and explicit empty identity.
M721	Extremum and arg-extremum	$\text{Seq}[A] \rightarrow \text{value/index}$	EXEC	Returns min/max values and stable indices/keys under a declared order, with empty and incomparable policies.
M722	Mean and weighted mean	$\text{Seq}[\text{number}] \times \text{weights?} \rightarrow \text{number}$	EXEC	Arithmetic or profile-defined means with weight validation, accumulator order and exactness status.
M723	Finite difference and recurrence step	$\text{Seq}[A]/\text{state} \rightarrow \text{Seq}[A]/\text{state}$	EXEC	Builds discrete derivatives, cumulative sums or recurrence transitions with explicit boundary and step policy.
M724	Function composition and pipeline	$(B \rightarrow C) \times (A \rightarrow B) \rightarrow (A \rightarrow C)$	EXEC	Creates a new operator by ordered composition, retaining intermediate type checks, effects and trace identities.
M725	Currying, partial application and bounded fixed point	operator $\times$ bindings/profile $\rightarrow$ operator/result	EXEC	Transforms multi-argument operators, binds selected arguments, or iterates to a fixed point under a declared convergence/budget contract.

EXEC = executable bounded reference mechanism or directly checked record contract. SPEC = typed advanced contract; full domain implementation is not claimed.

A.7 Vectors, matrices and tensors (M726-M741)

ID	Operator / mechanism	Typed signature	State	Normalized definition
M726	Vector construction and coordinates	scalars $\rightarrow$ Vector[n]	EXEC	Builds finite vectors with dimension, basis, coordinate frame, scalar field and unit metadata.
M727	Vector addition and scalar action	Vector x Vector or Scalar x Vector $\rightarrow$ Vector	EXEC	Module operations with dimension/frame compatibility and explicit scalar field.
M728	Dot and bilinear product	$V \times V \rightarrow$ Scalar	EXEC	Inner or general bilinear pairing with declared metric/form; Euclidean dot is one profile.
M729	Cross, wedge and oriented area product	$V \times V \rightarrow$ V or Multivector	EXEC	Three-dimensional cross or exterior product under a declared orientation and dimension; no universal cross product is assumed.
M730	Norm, seminorm and distance	$V \rightarrow$ nonnegative scalar	EXEC	Lp, weighted, matrix and user-defined norms plus induced distances with metric-law status.
M731	Normalize, project and reject	$V \times$ subspace $\rightarrow$ components	EXEC	Unit normalization and decomposition relative to a subspace/vector with zero and conditioning guards.
M732	Matrix construction, transpose and adjoint	Matrix[m,n] $\rightarrow$ Matrix[n,m]	EXEC	Builds shaped matrices and applies transpose, conjugate transpose or typed adjoint under an inner product.
M733	Matrix multiplication	Matrix[m,k] x Matrix[k,n] $\rightarrow$ Matrix[m,n]	EXEC	Ordered contraction over compatible inner dimensions; elementwise product remains a separate operator.
M734	Determinant and trace	Matrix[n,n] $\rightarrow$ Scalar	EXEC	Square-matrix invariants with exact/symbolic/numerical algorithm status and conditioning metadata.
M735	Inverse and linear solve	$A \times b \rightarrow x/\text{status}$	EXEC	Solves $Ax=b$ or forms an inverse only under nonsingularity/conditioning policy; solve is preferred over explicit inverse.
M736	Rank, nullspace and range	Matrix $\rightarrow$ rank/subspaces	EXEC	Exact or tolerance-profiled linear subspace queries returning basis plus numerical rank status.
M737	Eigenvalue and eigenspace family	Matrix[n,n] $\rightarrow$ spectral record	SPEC	Spectral decomposition with real/complex and defective-matrix status; eigenvectors retain normalization and ordering policy.
M738	SVD and pseudoinverse	Matrix $\rightarrow$ U,S,Vt/pseudoinverse	SPEC	Singular-value decomposition and Moore-Penrose pseudoinverse with tolerance, rank and error certificate.
M739	Tensor and Kronecker product	Tensor[A] x Tensor[B] $\rightarrow$ Tensor[A+B]	EXEC	Forms multilinear products with explicit axis ordering, basis and output shape.
M740	Tensor contraction and Einstein expression	tensors x index spec $\rightarrow$ tensor	SPEC	Contracts named axes or an explicit einsum-like index expression; repeated-index meaning is parsed and validated.
M741	Reshape, broadcast and axis permutation	Tensor x shape/permutation $\rightarrow$ Tensor	EXEC	Changes representation without changing element count unless a declared broadcast/replication rule applies; axis lineage is retained.

EXEC = executable bounded reference mechanism or directly checked record contract. SPEC = typed advanced contract; full domain implementation is not claimed.

A.8 Complex, quaternion and geometric algebra (M742-M757)

ID	Operator / mechanism	Typed signature	State	Normalized definition
M742	Complex construction and conjugation	real x real -> complex; complex -> complex	EXEC	Constructs $a+bi$ and applies conjugation under a declared complex field and precision profile.
M743	Complex magnitude and argument	complex -> nonnegative real/angle	EXEC	Returns modulus and principal or unwrapped phase with branch-cut and zero policy.
M744	Complex arithmetic family	complex operands -> complex	EXEC	Addition, multiplication, division, power, roots and elementary functions with explicit branch semantics.
M745	Euler and polar conversion	complex <-> (radius,phase)	EXEC	Converts between rectangular and polar form while preserving selected phase lineage for repeated windings.
M746	Quaternion construction and conjugation	R^4 -> Quaternion	EXEC	Constructs scalar-vector quaternion records and applies conjugation with component ordering fixed by schema.
M747	Hamilton quaternion product	Quaternion x Quaternion -> Quaternion	EXEC	Noncommutative quaternion multiplication with ordered operands and exact component formula.
M748	Quaternion norm, inverse and normalization	Quaternion -> Quaternion/status	EXEC	Norm-based inverse and unit normalization with zero/conditioning guards.
M749	Quaternion rotation and SLERP	Quaternion x Vector/time -> Vector/Quaternion	EXEC	Applies unit-quaternion sandwich rotation and shortest-arc interpolation with antipodal and near-equality policy.
M750	Dual-number forward derivative	Dual x Dual -> Dual	EXEC	Represents $a+b\epsilon$ with $\epsilon^2=0$ so arithmetic propagates first derivatives under supported operators.
M751	Dual quaternion rigid transform	Rigid transform <-> DualQuaternion	SPEC	Represents coupled rotation and translation using unit dual quaternions with normalization and composition order.
M752	Clifford algebra basis and grade	signature -> algebra type	SPEC	Declares signature (p,q,r), basis blades, grades and coefficient scalar field for a finite geometric algebra.
M753	Geometric product	Multivector x Multivector -> Multivector	SPEC	Associative Clifford product combining symmetric inner and antisymmetric outer parts under the declared metric.
M754	Exterior wedge product	Multivector x Multivector -> Multivector	SPEC	Antisymmetric grade-raising product with dimension and orientation metadata.
M755	Inner and contraction products	Multivector x Multivector -> Multivector/scalar	SPEC	Left/right contraction and scalar products with explicit grade convention to avoid notation ambiguity.
M756	Reverse, involution and Clifford conjugate	Multivector -> Multivector	SPEC	Grade-dependent sign transforms are distinct named operators and cannot be collapsed into one generic conjugation symbol.
M757	Versor and sandwich action	Versor x Multivector -> Multivector	SPEC	Applies ordered $v \times \text{reverse}(v)$ actions for reflections/rotations only when versor and invertibility guards hold.

EXEC = executable bounded reference mechanism or directly checked record contract. SPEC = typed advanced contract; full domain implementation is not claimed.

A.9 Differential calculus (M758-M773)

ID	Operator / mechanism	Typed signature	State	Normalized definition
M758	Limit	function x point/direction -> value/status	SPEC	Symbolic, analytic, interval or numerical limit with approach side, topology and convergence status.
M759	Single-variable derivative	$f:A \rightarrow B$ x point -> linear/local derivative	EXEC	Exact rule, automatic differentiation or bounded finite difference under a declared method and step/error profile.
M760	Partial derivative	$f:\mathbb{R}^n \rightarrow B$ x axis -> derivative	EXEC	Derivative with respect to one named coordinate while other coordinates and chart are explicit.
M761	Gradient	scalar field x metric -> vector field	EXEC	Metric-dependent vector corresponding to a scalar differential; coordinate covector and physical force remain separately typed.
M762	Jacobian	$f:\mathbb{R}^n \rightarrow \mathbb{R}^m$ -> Matrix[m,n]	EXEC	Matrix/linear map of first partial derivatives with input/output coordinate order fixed by schema.
M763	Hessian	scalar field -> Matrix[n,n]	EXEC	Second derivative bilinear form or coordinate matrix with symmetry and differentiability status.
M764	Divergence	vector field -> scalar field	SPEC	Trace of a covariant/Jacobian derivative under a declared volume/metric structure.
M765	Curl and rotational derivative	vector/one-form field -> field	SPEC	Three-dimensional curl or exterior-calculus analogue with orientation and metric conventions.
M766	Laplacian and Laplace-Beltrami	field -> field	SPEC	Scalar/vector Laplacian under Euclidean or manifold metric with boundary conditions separately declared.
M767	Directional derivative	f x direction -> derivative	EXEC	Derivative along a declared tangent/vector field, normalized only when requested.
M768	Lie derivative and bracket	field x tensor/field -> tensor/field	SPEC	Differentiates tensors along a flow and defines vector-field commutator with chart and regularity contracts.
M769	Exterior derivative	k-form -> (k+1)-form	SPEC	Grade-raising differential on forms satisfying $d(d(\omega))=0$ under a declared smooth structure.
M770	Differential and pushforward	map x tangent vector -> tangent vector	SPEC	Maps tangent vectors through a differentiable function; source and target tangent spaces are explicit.
M771	Pullback	map x target covariant object -> source object	SPEC	Transports covectors/forms/tensors contravariantly through a map with composition order retained.
M772	Covariant derivative and connection	connection x fields -> tensor	SPEC	Differentiates fields using an explicit connection; Christoffel data and coordinate transformation rules are part of the profile.
M773	Forward and reverse automatic differentiation	operator graph x seed/cotangent -> derivative trace	EXEC	Evaluates primal values with tangent or adjoint traces; mutation/effects and unsupported primitives require explicit custom rules.

EXEC = executable bounded reference mechanism or directly checked record contract. SPEC = typed advanced contract; full domain implementation is not claimed.

A.10 Integration, measure and transforms (M774-M789)

ID	Operator / mechanism	Typed signature	State	Normalized definition
M774	Definite and indefinite integral	$f \times \text{domain} \rightarrow \text{value/function}$	EXEC	Analytic antiderivative or bounded numerical integral with interval, orientation, singularity and error status.
M775	Line integral	$\text{field} \times \text{curve} \rightarrow \text{scalar}$	SPEC	Integrates a scalar or differential form along an oriented parameterized curve with explicit frame and domain.
M776	Surface and volume integral	$\text{field} \times \text{manifold/domain} \rightarrow \text{scalar}$	SPEC	Integrates over oriented parameterized/implicit domains with measure/Jacobian and boundary contract.
M777	Measure and measurable-set operator	$\text{measure} \times \text{set} \rightarrow \text{nonnegative extended scalar}$	SPEC	Declares a measure space and evaluates measurable sets/functions; finite counting and weighted measures are executable subsets.
M778	Quadrature and adaptive summation	$f \times \text{interval} \times \text{profile} \rightarrow \text{estimate}$	EXEC	Trapezoid, Simpson, Gaussian or adaptive rules with sample budget and error estimate.
M779	Convolution	$f \times g \rightarrow h$	EXEC	Continuous, discrete, circular or group convolution with boundary, normalization and algorithm policy.
M780	Correlation	$f \times g \rightarrow \text{correlation}$	EXEC	Cross/auto-correlation with conjugation, lag, normalization and finite-boundary policy.
M781	Fourier transform	$\text{signal} \rightarrow \text{spectrum}$	SPEC	Continuous/discrete Fourier transform with sign, normalization, sampling and frequency-order conventions.
M782	Inverse Fourier transform	$\text{spectrum} \rightarrow \text{signal}$	SPEC	Inverse transform paired with a declared forward convention and reconstruction/error status.
M783	Laplace transform	$\text{time function} \rightarrow \text{complex function}$	SPEC	One/two-sided transform with region-of-convergence metadata and inverse capability status.
M784	Z transform	$\text{sequence} \rightarrow \text{complex formal function}$	SPEC	Discrete-time transform with bilateral/unilateral and region-of-convergence profile.
M785	Wavelet transform	$\text{signal} \rightarrow \text{coefficient hierarchy}$	SPEC	Continuous/discrete multiscale analysis with wavelet family, boundary and level budget.
M786	Hilbert transform and analytic signal	$\text{signal} \rightarrow \text{signal/complex signal}$	SPEC	Principal-value transform and analytic-signal construction with finite signal/filter approximation status.
M787	Radon and projection transform	$\text{field} \rightarrow \text{projection family}$	SPEC	Integrates over hyperplanes/curves with sampling and reconstruction policy; projection data is not object identity.
M788	Mellin and scale transform	$\text{positive-domain function} \rightarrow \text{complex function}$	SPEC	Multiplicative-scale transform with convergence strip and normalization metadata.
M789	Integral theorem certificate	$\text{domain} \times \text{field} \times \text{hypotheses} \rightarrow \text{equality certificate}$	SPEC	Green, Stokes and divergence theorem records relate interior and boundary integrals only when smoothness, orientation and domain hypotheses pass.

EXEC = executable bounded reference mechanism or directly checked record contract. SPEC = typed advanced contract; full domain implementation is not claimed.

A.11 Probability, statistics and information (M790-M805)

ID	Operator / mechanism	Typed signature	State	Normalized definition
M790	Probability space and event	space x measure -> probability model	EXEC	Declares sample space, sigma-algebra/profile and normalized measure; finite exact spaces are the reference executable subset.
M791	Random variable and distribution	sample -> value; parameters -> distribution	EXEC	A measurable map plus named distribution parameters, support and seed/source policy.
M792	PMF, PDF and CDF family	distribution x value -> probability/density	EXEC	Probability mass/density and cumulative operators with normalization, support and mixed-distribution status.
M793	Expectation	distribution x variable -> value	EXEC	Finite sum, integral or estimator of a random variable with exactness and sampling error status.
M794	Variance and covariance	variables/distribution -> scalar/matrix	EXEC	Second central moment and cross-moment with sample/population and numerical-stability profile.
M795	Moment and cumulant family	distribution x order -> value/status	SPEC	Raw, central and standardized moments plus cumulants with existence and estimator status.
M796	Conditional probability and expectation	model x condition -> conditional model/value	EXEC	Conditions on an event/sigma-field with zero-probability and regular-conditional policy visible.
M797	Bayesian update	prior x likelihood x data -> posterior	EXEC	Combines prior and likelihood into normalized posterior with evidence/zero-normalizer and approximation status.
M798	Independence and dependence measure	variables/model -> status/measure	EXEC	Declares/checks independence in finite models and computes correlation or other selected dependence diagnostics.
M799	Seeded sampling and random stream	distribution x seed/state -> sample/state	EXEC	Deterministic pseudorandom or externally random sampling with algorithm, seed, stream ID and reproducibility status; randomness is not authority by itself.
M800	Likelihood and log-likelihood	model x parameters x data -> scalar	EXEC	Evaluates parameterized model evidence for observed data with support and underflow-safe log profile.
M801	Estimator and confidence interval	data x model/profile -> estimate	EXEC	Point/interval estimators with assumptions, finite-sample/asymptotic status and confidence interpretation.
M802	Hypothesis test and p-value record	data x hypotheses -> test record	SPEC	Test statistic, null/alternative, rejection rule, effect size and multiplicity policy are explicit; p-values are not posterior probabilities.
M803	Entropy and cross-entropy	distribution(s) -> scalar	EXEC	Shannon or declared entropy families over normalized distributions with log base and zero convention.
M804	KL and Jensen-Shannon divergence	P x Q -> nonnegative extended scalar	EXEC	Directed or symmetrized information divergence with absolute-continuity and zero-mass policy.
M805	Mutual information	joint distribution -> scalar	EXEC	Information shared under a declared joint distribution; geometric overlap alone is not substituted for a distributional calculation.

EXEC = executable bounded reference mechanism or directly checked record contract. SPEC = typed advanced contract; full domain implementation is not claimed.

A.12 Optimization, numerical methods and constraints (M806-M821)

ID	Operator / mechanism	Typed signature	State	Normalized definition
M806	Objective and feasible-set record	model -> optimization problem	EXEC	Declares decision variables, objective direction, constraints, domains, scales and evaluation budget.
M807	Argmin and argmax	function x domain -> optimizer record	EXEC	Returns optimizer set/value under exact, finite or approximate search with tie and nonattainment status.
M808	Gradient descent family	objective x initial x profile -> iterate/status	EXEC	Fixed, scheduled, momentum or adaptive first-order iteration with step, stopping and divergence contracts.
M809	Newton and quasi-Newton family	objective/system x initial -> iterate/status	SPEC	Uses Hessian/Jacobian or secant approximations with conditioning, damping and line-search safeguards.
M810	Line search and trust region	model x direction -> step/status	SPEC	Selects accepted steps under sufficient-decrease/curvature or local-model radius rules and finite evaluation budget.
M811	Conjugate gradient and Krylov iteration	A,b,initial -> x/status	SPEC	Solves compatible linear systems using matrix-vector products with symmetry/positive-definite and residual contracts.
M812	Linear programming	LP problem -> solution/certificate	SPEC	Linear objective and constraints with primal/dual feasibility, boundedness and certificate status.
M813	Quadratic programming	QP problem -> solution/status	SPEC	Quadratic objective with linear constraints and convexity/indefiniteness status.
M814	Nonlinear constraint and KKT record	problem x candidate -> KKT status	SPEC	Equality/inequality constraints, gradients, multipliers, complementarity and qualification status are explicit.
M815	Lagrange multiplier and penalty family	objective x constraints -> transformed problem	SPEC	Builds constrained objectives through multipliers, augmented Lagrangians, barriers or penalties with parameter schedule.
M816	Root finding	f x bracket/initial -> root record	EXEC	Bisection, secant, Newton or bracketed hybrid methods with interval, residual, derivative and termination certificate.
M817	Interpolation and approximation	samples x query -> estimate	EXEC	Piecewise linear, polynomial, spline or rational interpolation with node order, extrapolation and error/status contract.
M818	Least squares and regression	design x observations -> model	EXEC	Linear/nonlinear fitting with weights, regularization, residuals, rank and validation status.
M819	ODE and time-step solver	flow x state x time interval -> trajectory/status	EXEC	Explicit/implicit single or multistep integration with local error, stability, event and step-budget profile.
M820	Interval arithmetic and certification	interval operands -> enclosing interval	EXEC	Outward-bounded interval operations and relation classification prevent approximate values from crossing a guard unnoticed.
M821	Fixed-point iteration and convergence test	F x initial x profile -> fixed-point status	EXEC	Iterates $x_{k+1}=F(x_k)$ with norm, contraction/monotonicity evidence, cycle detection and hard budget.

EXEC = executable bounded reference mechanism or directly checked record contract. SPEC = typed advanced contract; full domain implementation is not claimed.

A.13 Geometry, topology and graph operators (M822-M837)

ID	Operator / mechanism	Typed signature	State	Normalized definition
M822	Affine transform	Affine x point/vector -> point/vector	EXEC	Applies $x \rightarrow Ax + b$ with coordinate-frame, composition order and invertibility status.
M823	Rigid, similarity and projective transform	transform x geometric object -> object	SPEC	Typed SE/Sim/projective maps with homogeneous-coordinate, normalization and orientation policy.
M824	Metric and distance relation	point x point -> nonnegative scalar	EXEC	Declares a metric, pseudometric or general dissimilarity with executable law status and unit.
M825	Implicit field and signed-distance relation	point -> residual/status	EXEC	Scalar inside/boundary/outside relation with exact-distance capability separately declared from generic implicit status.
M826	Constructive solid geometry	fields -> field	EXEC	Union, intersection, difference and bounded smooth blends over field signs with exactness/topology status.
M827	Curve and surface parameterization	parameters -> point	EXEC	Maps parameter domains to ambient coordinates with derivative, regularity, periodicity and frame policy.
M828	Curvature, torsion and shape operator	curve/surface x parameter -> geometric invariants	SPEC	Local differential geometry with regularity, frame and numerical conditioning status.
M829	Manifold chart and transition map	chart x point -> coordinates; charts -> transition	SPEC	Stores local coordinates, overlap domains and cocycle-compatible transition functions; coordinates are not identity.
M830	Boundary, coboundary and incidence	chain/cochain -> adjacent degree	EXEC	Cell/simplicial boundary and cochain coboundary maps with boundary-of-boundary zero validation.
M831	Homology, cohomology and persistence descriptor	complex/filtration -> invariant record	SPEC	Bounded algebraic-topology summaries with coefficient ring, filtration and solver/certification status; they do not replace geometry or lineage.
M832	Quotient, gluing and orientation transport	boundary event -> mapped state	EXEC	Explicit port identification, coordinate map, sheet/orientation update and lineage patch; projected intersections are not automatic collisions.
M833	Graph, hypergraph and adjacency construction	records -> graph	EXEC	Stable node/edge identities, directedness, weights, labels and incidence are explicit.
M834	Traversal, reachability and connected components	graph x start/policy -> ordered reachability	EXEC	Deterministic BFS/DFS and component queries with neighbor ordering, support and compatibility filters.
M835	Shortest path and path algebra	graph x endpoints -> path/status	EXEC	Dijkstra/Bellman-Ford/A* or semiring path evaluation with nonnegative/negative-cycle and heuristic contracts.
M836	Flow, cut and matching family	graph problem -> solution/status	SPEC	Network flow, cut, bipartite matching and related finite combinatorial optimization with capacity/weight certificates.
M837	Graph Laplacian and spectral operator	graph -> matrix/spectrum	SPEC	Combinatorial/normalized Laplacian and spectral descriptors with ordering, multiplicity and numerical status.

EXEC = executable bounded reference mechanism or directly checked record contract. SPEC = typed advanced contract; full domain implementation is not claimed.

A.14 Dynamics, control and event operators (M838-M853)

ID	Operator / mechanism	Typed signature	State	Normalized definition
M838	State-space model	(state,input,time) -> derivative/next/output	EXEC	Typed state, input, output, parameters and time base for continuous or discrete dynamics.
M839	Continuous ODE flow	flow x state x time -> state/status	EXEC	Maps initial state through a declared differential flow analytically or with a bounded solver.
M840	PDE operator and boundary conditions	field -> field evolution/status	SPEC	Spatial-temporal differential operator with domain, boundary/initial conditions, discretization and stability profile.
M841	Discrete map and recurrence	state x input -> next state	EXEC	State update $x_{[k+1]}=F(x_k,u_k)$ with tick, seed, branch and lineage semantics.
M842	Symplectic and structure-preserving step	Hamiltonian state x dt -> state	EXEC	Integration profile intended to preserve selected geometric structure qualitatively/approximately under a declared step and error budget.
M843	Finite-state machine	mode x input -> mode/output	EXEC	Finite modes, inputs, guards, transitions and outputs with deterministic priority and no hidden fallthrough.
M844	Hybrid automaton	hybrid state x time/input -> state/events	EXEC	Continuous mode flows plus guarded discrete resets, priorities and invariant regions.
M845	Guard relation	state/time -> guard status	EXEC	Scalar/Boolean/interval relation defines crossing, touch, tangency, coincidence or ambiguous status under a declared margin.
M846	Earliest event and root solver	trajectory x guards x horizon -> event/status	EXEC	Finds the earliest certified guard event in a horizon after support and compatibility, with equal-time interval handling.
M847	Support predicate	candidate x query -> admission	EXEC	Finite local geometric, temporal, semantic or resource domain prunes irrelevant candidates before expensive relation work.
M848	Compatibility predicate	state pair/context -> compatibility	EXEC	Typed mode, sheet, phase, orientation, ownership, policy, unit and schema conditions determine whether supported states may couple.
M849	Event proposal record	detector -> proposal	EXEC	Candidate change contains ID, time/interval, type, source, priority, payload, guard status, error, confidence and lineage label.
M850	Deterministic conflict resolution	proposal set -> accepted/rejected records	EXEC	Verified proposals are stably ordered; incompatible same-target patches are merged only by a versioned commutative policy or explicitly rejected.
M851	Atomic transition and reset	state x event -> new state	EXEC	One verified patch updates state, route, topology or mode atomically and preserves declared invariants.
M852	Checkpoint, replay and divergence detection	checkpoint x events -> state/divergence	EXEC	Canonical snapshots plus ordered events reconstruct state and stop at the first pre/post hash mismatch.
M853	Lineage and irreducible novelty	event/patch -> lineage append	EXEC	Persistent identity records seed/generative address, ordered transitions, external inputs and novelty that cannot be reconstructed from closed rules.

EXEC = executable bounded reference mechanism or directly checked record contract. SPEC = typed advanced contract; full domain implementation is not claimed.

A.15 Symbolic rewrite, proof and metaprogramming (M854-M869)

ID	Operator / mechanism	Typed signature	State	Normalized definition
M854	Symbol, binding and lexical scope	environment x symbol -> binding	EXEC	Names bind to typed literals, variables or definitions in explicit scopes; capture and shadowing policies are serialized.
M855	Substitution	term x substitution -> term	SPEC	Capture-avoiding replacement of free variables or pattern parameters with type and scope checks.
M856	Alpha conversion	term x fresh names -> equivalent term	SPEC	Renames bound variables while preserving binding structure and semantic identity.
M857	Beta reduction and application	function term x argument -> term	SPEC	Applies a function/lambda to arguments by capture-avoiding substitution under evaluation strategy and budget.
M858	Simplification	expression x law set -> expression	EXEC	Applies scoped exact identities and neutral/absorbing rules without changing domain, branch or floating-order semantics.
M859	Canonicalization	expression -> canonical expression	EXEC	Produces deterministic syntax/JSON form; commutative operand sorting occurs only when the type/law/exactness profile permits it.
M860	Pattern matching	pattern x term -> bindings/status	SPEC	Typed structural match with wildcards, guards, sequence variables and deterministic first/best-match policy.
M861	Rewrite rule	term x rule -> term/status	EXEC	Directed equation with preconditions, priority, phase, termination hints and provenance; bidirectional equations are not silently oriented.
M862	Term ordering	terms x order -> comparison	SPEC	Lexicographic path, degree/grade or custom well-founded order selects deterministic rewrite orientation and completion priorities.
M863	Common-subexpression elimination	expression DAG -> shared DAG	EXEC	Identical pure content-addressed subgraphs share one node; effectful or context-dependent expressions are not merged.
M864	Differentiation and integration rule registry	operator x rule context -> transformed operator	SPEC	Operator definitions may expose symbolic derivative/antiderivative templates with branch and domain guards.
M865	Equation and inequality solving	relations x variables -> solution record	SPEC	Returns exact, parameterized, interval or numerical solution sets with assumptions and completeness status.
M866	Assumption and theorem context	context x statement -> scoped statement	SPEC	Stores hypotheses, axioms, definitions and imported certificates separately from conclusions and computational evidence.
M867	Proof obligation and certificate	claim x context -> obligation/certificate	EXEC	A transformation may emit law, type, domain, error or invariant obligations; discharge status is machine-readable and never inferred from confident prose.
M868	Bounded recursion and grammar	seed x productions x budgets -> finite graph/status	EXEC	Recursive definitions require depth, symbol, stack, time or memory budgets and explicit cycle/fixed-point policy.
M869	Operator generator and closure schema	generator x definitions -> operator family	EXEC	Meta-operators construct identity, constant, projection, composition, product, map, fold, derivative, adjoint, inverse and user-defined families, giving finite representation to open-ended operator classes.

EXEC = executable bounded reference mechanism or directly checked record contract. SPEC = typed advanced contract; full domain implementation is not claimed.

A.16 Determinism, compression, exchange and UGTS handoff (M870-M885)

ID	Operator / mechanism	Typed signature	State	Normalized definition
M870	Canonical JSON and binary serialization	record x profile -> bytes	EXEC	Stable key order, number spelling, UTF-8 and typed binary profiles make records reproducible; NaN and implementation-specific objects require explicit encoding.
M871	Definition and expression content hash	record -> digest	EXEC	SHA-256 over canonical records verifies identity and cache integrity; hashes are not semantic proofs or legal signatures.
M872	DAG structural sharing	graph -> deduplicated graph	EXEC	Pure identical definitions and expressions are stored once and referenced by content address, preserving order and dependency lineage.
M873	Seed plus parameter reconstruction	seed x graph x params -> reconstructed state	EXEC	Closed deterministic families may store seed, versioned operator graph and parameters instead of materialized samples; external novelty and checkpoints remain authoritative.
M874	Opcode and codebook encoding	operator IDs <-> compact words	EXEC	Common operator IDs map to a versioned compact codebook; unknown codes reject and codebook changes require migration.
M875	Exact round-trip contract	record -> bytes -> record	EXEC	Encode/decode must preserve canonical operator graph, order profile, parameters and required error metadata; size reduction alone is not compression proof.
M876	Quantization profile	value x profile -> code/interval	EXEC	Finite-width encoding declares range, step, saturation, special values and reconstruction error for every field.
M877	Interval and error propagation	input intervals x operator -> output interval	EXEC	Operator-specific enclosures propagate uncertainty so support, compatibility and guard decisions can reject ambiguous results.
M878	Stable floating numerical profile	runtime -> numerical capability record	EXEC	Declares precision, rounding, fused-operation, signed-zero, denormal, transcendental and tolerance behavior; universal cross-platform bit identity is not assumed.
M879	Parallel schedule and deterministic join	DAG x scheduler -> deterministic trace	EXEC	Independent pure nodes may execute in parallel, but commits and reductions use recorded stable join/order policies and conflict checks.
M880	Capability and profile negotiation	capability sets -> compatible profile/status	EXEC	Peers or adapters exchange schema, operator IDs, exactness, order, precision and budget support before accepting an expression or replay.
M881	Schema and catalog validation	package records -> validation report	EXEC	Validates required fields, unique IDs, contiguous merged mechanism range, content hashes, dependencies, precedence and example references.
M882	Trace and explanation query	evaluation -> deterministic trace	EXEC	Returns parse tree, resolved operators, types, phases, operand/result values, error status, proposal reasons and content hashes.
M883	Cost, memory and budget record	operator/profile/run -> resource record	EXEC	Operators declare asymptotic class, bounded work, sample/iteration/depth limits and observed resource metrics separately from semantic output.
M884	Kill criteria and fallback policy	evaluation/profile -> continue/delegate/reject	EXEC	Rejects or delegates when type/law/order is ambiguous, error exceeds margin, budgets overflow, performance loses at equal error or a conventional method is simpler.
M885	Canonical UGTS operator handoff	operator result -> UGTS event sequence	EXEC	Pure mathematical results and certificates feed local support, compatibility, guard classification, verified proposal, deterministic commit, transition and lineage; rendering and I/O remain downstream.

EXEC = executable bounded reference mechanism or directly checked record contract. SPEC = typed advanced contract; full domain implementation is not claimed.

Appendix B. Precedence Profiles and Phase Schedule

B.1 Standard mathematical precedence

Binding power	Level	Associativity
100	delimiters_application_indexing	left
90	postfix_factorial_transpose	left
80	power_root	right
70	unary_sign_bitnot	right
60	multiplication_division_modulo_contraction	left
50	addition_subtraction	left
40	set_relation_comparison_chain	nonassociative_or_chain
35	logical_not	right
30	logical_and	left
25	logical_xor	left
20	logical_or	left
15	implication_equivalence	right
10	conditional_case	right
5	definition_assignment	right

B.2 Other profiles

Profile	Purpose	Implicit multiplication
standard-math-4.2	Standard mathematical token profile.	False
python-safe-subset-4.2	Uses the Python expression AST only as a safe syntactic carrier; calls are restricted to the bundled registry and no attribute access, comprehensions or arbitrary code execution is allowed.	False
explicit-ast-4.2	No precedence dependence: every expression is a JSON node with operator_id and ordered argument list. This is the canonical exchange profile.	False

B.3 Canonical phase schedule

Index	Phase	Purpose
0	parse	Parse literals, symbols and explicit grouping into an AST or load canonical AST.
1	normalize	Normalize numbers, units, sign conventions, profiles and symbol aliases.
2	resolve	Resolve operator/definition IDs, verify content hashes and reject missing references or undeclared cycles.
3	type	Infer/check sorts, shapes, units, arity, domains and effect classes.
4	rewrite	Apply only scoped exact rewrites and canonicalization permitted by declared laws.
5	plan	Topologically order dependencies, choose kernels/algorithms and record reduction/composition order.
6	evaluate	Evaluate pure exact and bounded approximate operator nodes with a deterministic trace.
7	certify	Propagate intervals/error, discharge available proof obligations and classify numerical status.
8	support	Prune by finite local geometric, temporal, semantic or resource support.
9	compatibility	Check sheet, phase, mode, orientation, ownership, units, schema and application policy.

Index	Phase	Purpose
10	guard	Classify crossing, touch, tangency, coincidence, interior, exterior or ambiguous relation status.
11	proposal	Emit a typed candidate event or patch without mutating authority.
12	commit	Deterministically order verified proposals, resolve conflicts and apply one atomic patch.
13	lineage	Append pre/post hashes, transition identity, external novelty and replay information.
14	projection	Optionally render, export, store or actuate downstream; projection never becomes mathematical or event authority.

Appendix C. Package, Source and Attribution Register

C.1 Final package map

Path	Contents
report/	PDF and editable DOCX source.
spec/	Formal definition, 256-entry CSV/JSON catalog, schema, order profiles, phases and source register.
src/ugts_kc42/	Dependency-free reference runtime and bundled data.
examples/	Literal content-addressed project and deterministic demonstration output.
tests/	441-test suite.
docs/	API, order profiles, migration, boundaries and reproduction.
figures/	Original report diagrams.
dist/	Wheel and source archive.
validation/	Test, schema, demo, build, clean-install and PDF/DOCX evidence.
manifest/ / checksums/	Final inventory, package tree and SHA-256 release list.

C.2 Source register

Source	Role	SHA-256
UGTS-KC 3.9 KC Elizabeth Vector Game Runtime	deterministic game runtime, ordered phases, projects and validation	d7ee3c91abf60e35b4b15fb296477974e45802c24e1fde618d683949bc1c7df2
UGTS-KC 3.6.2 SCLP	typed symbol separation, interval guards, finite indexing and evidence boundaries	6987490a0a7b7ed76b0091057a2e69ab831fee786434a0b4219782923f5ea580
UGTS-KC 2.0 Expanded Substrate	patterns, fields, topology, kinematics, dynamics and event degeneracy	a29dd9959ebd300c32d0e1df8fa2c0d4ee15f428892557b98b5f18cd3e878399
UGTS-KC Two Hands 3.0	stable scene identity, proposal verification, deterministic commit and replay	183ffa4e2bf29cfa0308687cd221529321598dbd69ef75c150abc0c64810f214
UGTS-GN 1.1 GPU-native Addendum	canonical authority chain, packed precision, compression and hardware evidence rules	3a6b6c57118ca4e2d7d7e577508c3477a095d133791f3708cb56a054285225d2
UGTS-KC 3.6 Literal Referential Substrate	content-addressed definitions, dependency DAG and normative operation order	e21433cd8378368d4fe4ae278d3c9c364725e45ae7fd75b743328bd2d6003a46
UGTS-KC 3.6.3 SARA	typed security/public boundaries and authorization guards	c5a8256e345047d56fbd56b7ce2ba77d95dd8391fc91bf6528295831714bab2f

C.3 Project continuity records

Record	4.2 continuity use	Availability
UGTS-KC 3.9.3 IQ Field	catalog continuity through M609 and bounded procedural/query operators	project context; not redistributed

Record	4.2 continuity use	Availability
UGTS-KC 3.9.4 Bayer Direct Native	catalog continuity M610-M629 and downstream finite display projection	project context; not redistributed
UGTS-KC 4.0 Spatial Evidence Ledger	typed observation uncertainty, stable map identity, ledger and replay	project context; not redistributed
UGTS-KC 4.1 POCO X7 Pro Spatial Seed Native	seed/evidence storage and named-device promotion boundaries	project context; not redistributed

C.4 Attribution notice

Prepared for Tom Klootwijk as the UGTS-KC 4.2 General Operator and Order Addendum. The requester-supplied identifier NL200678942 and date of birth 10-07-1990 are recorded only as supplied and were not independently verified. This report and package are not legal proof of identity, authorship, ownership, patentability, priority, licensing status or chain of title. No third-party source is silently relicensed by this wording.

C.5 Release identifiers

Identifier	Value
Runtime version	4.2.0
Codename	GOOA - General Operator and Order Addendum
Project schema	ugts-kc-general-operator-order-4.2
New mechanism range	M630-M885
Merged catalog spine	M001-M885 by project/source/catalog continuity
Operator records	256
Automated tests	441
Demo content hash	sha256:abc476fd1121a63ceed9d489097c4864b6f64ac9fdce1cfce2d264fab4908fe

CLOSING BOUNDARY

This package establishes a coherent, inspectable and reproducible general operator/order reference design. It does not establish mathematical closure, universal solver coverage, production safety, physical-device performance or legal ownership.